SuperAdmin Manual

FinOps SaaS · CSCloudSolutions

Version 2.1 · October 2026

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📘 SuperAdmin Manual — FinOps SaaS (CSCloudSolutions)

Version: 2.1 (detailed) **Language:** English **Last updated:** October 2026 **Audience:** CSCloudSolutions internal team only (SuperAdmin). Includes everything in the User Manual plus platform administration flows (tenant onboarding, customer onboarding, sales tracking).

Recent updates (September – October 2026)

Everything that changed for a tenant's user is in the **User Manual** and is included here in sections 3 to 12 (Perspectives and team scope, Leaderboard and gross margin, report subscriptions, fiscal year and budget rollover, end of support, Shift-Left, Cosmos DB, egress and link capacity, Azure Virtual Desktop, Storage, Marketplace, mobile app). What changed on the platform-operations side:

- **Affiliate Program with its own partner panel (17):** the affiliate enters with a personal read-only link, without an account in any tenant; a referral's commission lasts **the first year** of each client (12 months counted from their first charge) and not forever.
- **Payout runs (17.6):** the due commissions of affiliates and sales reps are settled in batches per currency, with a CSV payment order and confirmation with a receipt.
- **Sales commissions (20):** sale date and contract term per tenant, and the sales rep's accrual schedule (annual or monthly) on the same ledger as affiliates.
- **Platform usage (22):** AI tokens absorbed by the house and calls to Azure APIs, with the throttling percentage per service and per tenant.
- **Signup funnel (21), Pricing units (23), Load testing (24) and AI — Global configuration (25):** documented for the first time in this manual.
- **Anomalies with an owner and follow-up (18) and subscriptions beyond a plan's limit (19):** what the client sees and how to explain it.
- **Links that leave the platform:** the partner-panel link and the other links we email use the configured public URL (`NEXT_PUBLIC_APP_URL`), not the host the requester came in through.
- **Infrastructure:** the production `terraform apply` is done **only through CI** (`terraform.yml` workflow, `APPLY-PROD` confirmation), never from a local machine; production environment values live in the `TF_VARS_PROD` secret.

August 2026 updates

- **Compute FinOps Cockpits & Serverless Efficiency:** Added resolution cockpits for **Virtual Machines** (`/intelligence/computo/avm`) with active compute vs. persistent storage cost separation (deallocated waste detection), AHUB licensing, B-series rightsizing, 8x5 schedules, and abandoned VM disposal; **Function Apps** (`/intelligence/computo/fapps`) with invocation telemetry, GB-seconds, dependency audits (Storage/App Insights), and sampling/downgrade playbooks; **App Services** (`/intelligence/computo/waas`) with app density and packing; and **VMSS** (`/intelligence/computo/vmss`) with autoscale and Spot pricing.
- **Responsive Popovers Standardization (`InfoTooltip`):** 100% responsive popovers with institutional `#1B2A41` background ensuring zero text clipping across any viewport.
- **New SuperAdmin panel:** `Partner Center Alerts` (`/superadmin/partner-alerts`) to monitor PAL/CPOR status by tenant and detect recent events.
- **New Azure Integration Services (iPaaS) module:** `Intelligence → Azure Integration Services` now includes tabs for Logic Apps, APIM, Service Bus, Event Grid, Event Hubs, and ADF, with a dedicated Enterprise Connectors section for Logic Apps.
- **PAL operational automation:** when partner association is approved, events are logged and automatic retry (cron) updates status and detail.

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- **Enterprise AI (Azure IA):** global configuration now stores and tests **endpoint URL** + deployment in addition to API key, with backward-compatible fallback.
- **Cross-cutting FinOps/CMP table directive:** formalized for current and future tables (base filters, A-Z/Z-A/cost sorting, 15/30/45/60 pagination, responsive full-width layout, and resizable columns).
- **Security and Monitoring standardized:** both modules now match the same style used in Databases and Compute, including pagination and resource/region/type/resource-group filters.
- **New Security operational cron:** `prewarm-security-finops` added to Terraform cron maps (staging/prod) to prewarm `defender` and `security/service-cost`.
- **Key Vault Hardening & Network Isolation:** Production Key Vault protected with a dedicated Private Endpoint (`cscs-finops-prod-wus2-kv-pe`) in `snet-pe`, firewall set to `default_action = Deny`, and automated ephemeral runner opening in GitHub Actions.
- **Exhaustive Daily Prewarm (4:00 AM) & Async Poll Architecture:** Deployed `prewarm-daily` cron (`0 4 * * *`, 07:00 UTC) to pre-calculate and cache to Redis (26h TTL) full KQL audits, zombies, whiteboard, MTD/historical costs, inventory, tags, FinOps maturity, and scorecards. Migrated `prewarm-databases` and `prewarm-compute` Container App Jobs to `async_poll = true` protocol to eliminate 240s Ingress timeouts.
- **Azure SQL Elastic Pool Billing Correction:** Assigned `$0.00 (In Pool)` cost to Elastic Pool member databases, preventing duplicate multi-database compute multiplication and properly billing at the Elastic Pool resource level.
- **AI Cost Analytics hardening:** fixes for real-consumption display, daily MTD trend, and cache/data-source robustness for Microsoft Foundry.
- **Database metrics hardening:** Redis, MySQL, PostgreSQL, Cosmos DB, MongoDB, and SQL/Managed Instance now use per-metric fallback to prevent `N/A/unknown` dashboards under partial Azure telemetry.
- **Tabler-based FinOps visual refresh:** key modules in Intelligence, Consumption, Governance, Cleanup, Overview, and Copilot M365 are now standardized with no blue icon backgrounds.
- **Staging pipeline hardening:** deployment now resolves subscription/RG/app/job dynamically, tracks migrations with the proper `job-execution-name`, and treats redirect health checks (`200/307/308`) as healthy.

Change-control directive (mandatory)

For agent-assisted tasks in this repository:

- **Do not push to `staging` or `main` without explicit user instruction.**
- **Do not merge into `main` without explicit user instruction.**
- Default mode is **local commits only** until an explicit promotion request is provided.

How to use this manual

Each section explains **what** the feature is, **who** can use it (role and subscription tier), and the **step-by-step usage flow** — concrete buttons, URL paths, form fields, and expected results. If you're a non-technical user, feel free to jump straight to the section you need: each one is self-contained.

📑 Table of Contents

1. Getting Started
2. Roles and Permissions
3. Main Navigation
4. Visibility Section
5. Financial Intelligence Section

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6. Cloud Cleanup Section
7. Governance Section
8. Administration Section
9. FinOps Copilot (AI Assistant)
10. Account Security (MFA)
11. Advanced Features and Integrations
12. Best Practices
13. Global Communications
14. Tenant Lifecycle
15. Azure Marketplace
16. Onboarding via Azure Lighthouse (Enterprise)
17. Affiliate Program
18. Spend anomalies: who handles them
19. Subscriptions beyond your plan's limit
20. Sales commissions
21. Signup funnel
22. Platform usage
23. Pricing units
24. Load testing
25. AI — Global configuration

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1. Getting Started

1.1. Access and login

The platform is a B2B SaaS. Authentication is integrated with **Microsoft Entra ID** (Azure Active Directory):

1. Go to the platform URL.
2. Click **"Sign in with Microsoft"**.
3. Authenticate with your corporate account. The platform automatically recognizes your Azure tenant and identity.
4. **Demo Mode:** if you want to try the platform without connecting your real environment, choose one of the preconfigured demo profiles from the main screen — they come with realistic simulated data and metrics, so you can explore every module risk-free. There are four demo tenants (one per tier).

1.2. The onboarding wizard (first time)

If you're an administrator and this is your organization's first time using the platform, upon login the **Setup Wizard** (`/onboarding`) automatically opens — a 5-step linear wizard with a progress bar:

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Step	What you do	Result
1. Welcome & Company Info	Confirm company name, primary cloud provider (Azure), display currency (USD/EUR/GBP), and timezone	Your initial preferences are saved
2. Connect Azure Subscription	Paste **Client ID**, **Client Secret**, and **Azure Tenant ID** from the Service Principal (generated with the PowerShell script provided by CSCloudSolutions) and click **"Validate"**	The system checks live whether the Service Principal has the minimum required roles; if any is missing, it shows exactly which one in red
3. Run First Data Sync	Click **"Run Sync"**	Fetches your first set of cost data from Azure (can take up to 60 seconds)
4. Create Your First Budget	Fill in name, monthly limit ($), and alert threshold (%)	Your first active budget is created
5. Set Up Notifications	Click **"Configure"** (opens `/admin/notifications` in a new tab)	Add at least one channel (email, Slack, or Teams) to receive alerts

You can **skip** any step and return later — the wizard keeps appearing until you complete or skip all 5. Progress (%) is calculated as `(completed + skipped) / 5 × 100`. If you prefer the advanced setup form instead of the wizard, `/admin/onboarding` remains available in parallel.

1.3. Onboarding a new customer (SuperAdmin flow)

If you're a CSCloudSolutions SuperAdmin onboarding a new tenant:

1. **Register the Tenant:** go to `/admin/tenants` → **Tenant Management** → enter the customer's Entra ID, business name, and initial Tier.
2. **Generate credentials:** once created in the database, go to `/admin/onboarding` → **Customer Onboarding**. The **Client ID** and **Client Secret** fields now appear, to paste the Service Principal credentials generated by the PowerShell script you shared with the customer.
3. **Tag commercial origin and commission (optional):** in each tenant's expanded panel in the Environments Directory, set **"Commercial Origin / Referred by"** and **"Commission (%)"** for internal commission settlement — visible only to SuperAdmins.
4. **Partner association (PAL / CPOR):** after credentials are loaded, the approve/decline block remains visible until the state is **linked (LINKED)**. If it ended in `FAILED` or `DECLINED`, you can retry without manual state reset.
5. **Secure Customer Session Impersonation:** from the Tenants table (`/admin/tenants`), each row features an **"Impersonate"** action button (`IconUserCheck`). Clicking it:
   - Sets an encrypted HTTP-only cookie `saas_impersonation_session` with a 4-hour lifespan.
   - Grants the SuperAdmin immediate visibility into the exact tenant workspace, subscriptions, and RBAC views.
   - Displays a fixed top banner (`z-[90]`) with delegation details and an instant **"Exit Impersonation"** return button.
   - Automatically writes an immutable audit record in `AuditTrailLogs` with `isImpersonated: true` and `executedBySuperAdmin` .

Azure roles the script assigns, by contracted tier:

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Tier	Built-in roles	Custom role (*CSCloudSolutions Remediation Role*)
Professional (platform floor)	Reader, Cost Management Reader, Monitoring Reader, Billing Reader, Security Reader	— (read-only)
Business	Professional + Tag Contributor	Start, stop, restart and deallocate VMs; write tags; manage Azure budgets; and **delete** the orphaned resources that Cloud Cleanup detects (VMs, disks, snapshots, App Service plans, NICs, public IPs, NSGs and the other network types of Networking Zombies)
Enterprise	Business + Monitoring Contributor	Business + create resource groups

In addition, the script grants the identity **Microsoft Graph read permissions** on all plans (`Directory.Read.All`, `Reports.Read.All`, `User.Read.All` and `Organization.Read.All`) and, depending on the plan: `Application.ReadWrite.OwnedBy` (Business+, to rotate secrets: it only reaches the App Registrations the identity *owns*), `AuditLog.Read.All` (Enterprise, sign-in activity for *Users & Licenses*) and `ExternalConnection.ReadWrite.OwnedBy` (Enterprise, Copilot M365 connector).

> ⚠️ **Reader, Cost Management Reader, Monitoring Reader and Billing Reader are the absolute minimum** for the Real Consumption page to show data. If `Cost Management Reader` or `Billing Reader` is missing, Azure silently returns 0 rows. `Security Reader` adds read access to the security posture (Defender/WAF).

> ℹ️ **AI Cost Analytics (Microsoft Foundry / Azure OpenAI)** uses the same base roles (`Reader`, `Cost Management Reader`, `Monitoring Reader`, `Billing Reader`): **no extra role is required**.

> ⚠️ **EA/MCA subscriptions** (Enterprise Agreement / Microsoft Customer Agreement) additionally require the customer's Billing Admin to assign `Enrollment Reader` or `Billing Account Reader` to the Service Principal at the billing account scope — the script can't do this automatically, it must be coordinated with the customer.

Verifying permissions were correctly assigned, after the customer runs the script:

- **Quick option (recommended):** `GET /api/admin/check-sp-roles?tenantId=<tenant-id>` — returns how many subscriptions have all required roles ✅, which are partial ⚠️, and which have none ❌, with the exact detail of what's missing in each.
- **Manual option:** in Azure Portal → Subscription → **Access control (IAM)** → **Role assignments** → filter by the App Registration `CSCloudSolutions-FinOps-Agent` and confirm the tier's roles appear.

Troubleshooting — "Real Consumption shows no data":

Error message	Cause	Fix
`NO_COST_PERMISSION`	Missing `Cost Management Reader`	Assign the role to the SP on that subscription
`NO_SUBSCRIPTION_ACCESS`	Missing `Reader`	Assign `Reader` or re-run the script
`SUBSCRIPTION_INACTIVE`	The subscription has no spend in the current month or the last 30 days	Verify it's the correct subscription
`NO_SUBSCRIPTIONS`	The SP sees no subscriptions	Assign `Reader` on at least one
`NO_CONSUMPTION`	Everything OK but no spend in the current cycle	Wait for the billing cycle to close or check another subscription

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Troubleshooting — "AI Cost Analytics (Microsoft Foundry) shows 0 tokens / $0":

Even when models (e.g. gpt-5.1, gpt-5.3-codex) have real usage in Microsoft Foundry, the panel may stay at zero for several reasons. Use the **diagnostics endpoint** (no database or cron-log access required):

- `GET /api/intelligence/ai-analytics/diagnostics?tenantId=<tenant-id>` — returns, step by step: current rows in `AICostSnapshots`, visible subscriptions (with tier truncation), discovered `Microsoft.CognitiveServices/accounts` and their `kind`, the **metrics actually available** per account, a **live probe** of each token metric (series count + sum) and the rows the collector would produce, plus a **conclusion** pointing to the likely cause.

Diagnostics conclusion	Cause	Fix
`No visible subscriptions`	The SP lacks `Reader`, or the tier truncates subscriptions	Assign `Reader` / review tier
`No CognitiveServices account found`	The Foundry resource is a different type or in a subscription beyond the tier limit	Verify the Foundry resource type/subscription
`AI accounts do NOT expose token metrics`	Foundry models don't emit `ProcessedPromptTokens` / `GeneratedTokens` to Azure Monitor	Cost arrives via the `CostSnapshots` fallback (Cost Management, 8–24 h latency)
`Metrics exist but return 0`	Azure Monitor latency (~15 min) or usage outside the yesterday+today window	Retry in a few minutes and re-run the `/api/cron/sync` cron
`Collector produces rows`	All OK; just needs persisting	Run the `/api/cron/sync` cron (invalidates the cache automatically)

> ℹ️ **Fix 2026-08-05:** the collector now requests each token metric **separately** from Azure Monitor. It previously requested them in a single batch and, if one didn't exist for the resource (e.g. `ProcessedInferenceTokens` on classic Azure OpenAI), Azure rejected the **whole** batch with a 400 and the panel stayed at zero.

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2. Roles and Permissions

The platform separates **two independent concepts** that combine but don't replace each other:

2.1. Role (what you can DO)

Role	Capability
SuperAdmin	Full administration of CSCloudSolutions: create tenants, configure payment gateways. Internal role, not for customers.
Owner	Tenant owner. Full access, including plan changes and billing.
Admin	Complete financial visibility, configuration changes, corrective actions (shutting down VMs, deleting resources).
Colaborador (Collaborator)	Sees financial intelligence and visibility; can suggest changes but doesn't manage billing or users.
Reader	Read-only in dashboards and reports. Cannot apply changes or view sensitive configuration.

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2.2. Domain permissions (which pages you SEE)

Beyond the role, each user can be assigned one or more **domain permissions**, which determine which sidebar sections they see — independent of their role:

Permission	Who benefits	What it enables to view
FinOps	FinOps Analyst	Cost and recommendation pages
CloudAdmin	Cloud Administrator	Infrastructure execution/write pages
Security	Security Auditor	Compliance auditing, credentials, governance
ProductOwner	Project/product lead	Visibility by Cost Center or application

How they combine: a user with the **Reader** role and the **FinOps** permission can *view* FinOps pages, but can't modify or delete anything in them — the role still governs actions. Permission filtering is opt-in: if a user has no permission assigned, they see everything their role and tier allow (default behavior, no domain restriction). **Admin/Owner** roles always see everything their tier allows, regardless of assigned permissions. The **Dashboard** (`/`) and **Support** (`/support`) are always visible to everyone.

How to assign permissions: go to `/admin/users` → select the user → toggle the permission switches in the table, or assign them when manually creating a new user. (Note: users bulk-imported from Entra ID have no permissions assigned by default — assign them afterward from the main table.)

> ⚠️ Domain permissions **do not** enable write actions on the backend — that remains exclusively controlled by the role. A user with the `CloudAdmin` permission but `Reader` role can *view* those pages, but any write action will return a 403 error if their role doesn't authorize it.

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3. Main Navigation

The left sidebar groups everything into 5 pillars (collapsible sections — the active page's section auto-expands):

- **Visibility** — Whiteboard, FinOps Academy, Azure Advisor, FinOps Maturity, Historical Progress, TOP Expenses, Resources, Green FinOps, Captured Savings and Financial Leaks.
- **Financial Intelligence** — Consumption & Budget, Team leaderboard, Perspectives, Optimization & Savings, Databases, Compute, Storage, Networking, Users & Licenses, Azure AI, Azure Integration Services, Monitoring, Security and Advanced Analytics.
- **Cloud Cleanup** — Zombie Resources, Networking Zombies, TTL Expirations and Orphaned Backups.
- **Governance** — Tag Compliance, Self-Service Alerts, Power Schedules, Auto-Block Policies, Governance Reporting, High Availability, End of support, Expiring Credentials and Approvals.
- **Administration** — Support, Users & Access, Configuration, Reports & Export, Integrations & API, Billing & Audit and Marketplace.

What you see depends on three things: your **plan** (each screen states the plan it is available from), your **role** (what you can do) and your **domain permissions** (which pages are shown to you). Whatever your plan does not include appears with a lock and an upgrade notice, or can be added on its own from the **Marketplace** (8.10).

A **search bar** at the top of the sidebar lets you find pages by name without browsing the categories, and it only lists what you can open.

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3.1. "History" button — time evolution of any metric

On Dashboard, Commitment Discounts, Rightsizing, Anomalies, Budgets, and High Availability you'll find a **History** button in the top right corner. When opened:

1. Choose a date range (up to **1 year back**).
2. See the daily evolution of that page's metrics as a **line chart** and **table**.

The platform automatically saves a daily snapshot of every page (retention ~13 months) — you don't need to enable anything, it's already running in the background.

3.2. Your profile

Click your **avatar** (circle with your initial, top right) to open:

- **Full name** — editable via the pencil icon.
- **Email and role** within the tenant (read-only).
- **Display currency** — currency selector.
- **Appearance** — Light / Dark / Automatic (follows the OS theme).
- **Sign out.**

3.3. Hubs with tabs

Many menu entries are **hubs**: one screen with tabs that groups the views of a single topic. You enter through the hub and move between its tabs without going back to the menu; the active tab stays in the URL, so you can share the exact link. Hubs include **Consumption & Budget**, **Optimization & Savings**, **Databases**, **Compute**, **Storage**, **Networking**, **Monitoring**, **Security**, **Azure AI**, **Azure Integration Services**, **Advanced Analytics** and, in Administration, **Users & Access**, **Configuration**, **Reports & Export**, **Integrations & API** and **Billing & Audit**.

The URLs that predate the hubs (for example `/intelligence/billing` or `/admin/users`) still work: they open the same screen or redirect you to its tab, so old bookmarks do not break.

3.4. The scope bar: subscription, Perspective and Views

The header has three selectors that narrow what the dashboards show:

- **Scope (tenant and subscription).** You pick the whole tenant or one Azure subscription. If your organization has more than one tenant, you also switch tenants from here.
- **Perspective and group.** It appears only if your plan is Enterprise and the tenant has Perspectives defined (5.3). You pick a Perspective and a group —for example *Teams* → *Engineering*— and the dashboards that support it show only that slice of the spend. The choice travels in the URL: it can be shared by link and survives a refresh.
- **Views** (bookmark icon). With a Perspective selected, *Save the current view…* remembers the combination — Perspective and group— under the name you give it, and you apply it again in one click from the same selector. Views you save from the screen are personal. Views created through the API (`/api/intelligence/views`) can also be **shared** —the whole tenant sees them, and only Admins and Owners create or delete them— or **default**, applied automatically on entry when the URL does not specify a scope. The selector is shown only when there are saved views or an active Perspective to save.

> If your account has a **scope assigned** by an Admin (8.2), the Perspective selector is fixed on your group, with a lock, and the Views selector is not shown: your slice is not chosen, saved or changed by you.

3.5. Shortcuts and mobile app

- **Command palette:** `⌘K` (Mac) or `Ctrl+K` (Windows/Linux) opens a search for screens, quick actions and also resources in your environment (type at least two characters).

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- **Manual:** the help icon (**?**) in the header opens this manual as a PDF, in your active language.
- **Phone:** on narrow screens the platform offers a dedicated mobile experience (11.12).

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4. Visibility Section

4.1. Dashboard (White Board)

Your executive entry panel. It brings together Current Cost, forecast, Zombie Resources, Potential Savings and Environmental Impact, along with budgets, dominant services, governance, Advisor and Quick Wins.

How to use it:

1. Choose the **period** in the header selector: **Current Month (MTD)**, **Last 30 days** or **Last 90 days**. The current month is queried live from Azure Cost Management (with the platform's database as a fallback if Azure does not respond); the 30- and 90-day ranges are read from the history the platform stores every night, so they answer instantly and do not spend Azure query quota. The main cost card changes its label according to the range.
2. Check the sync bar and use **Refresh** to bypass the cache and query the current cycle.
3. Compare spend against the cost-center budgets and adjust the 12-month forecast rate.
4. Pin/unpin widgets from the pin icon to take them to **My Dashboard**.
5. In **Top Quick Wins**, choose **Optimize** to open the remediation modal with CLI, PowerShell and Terraform scripts.

The Whiteboard respects the **scope** of the top bar (3.4): with a Perspective and a group selected, the numbers are those of that slice of the spend.

4.2. Spend & Forecast (tab of the Consumption & Budget hub, Enterprise tier)

This screen lives as a tab of the **Consumption & Budget** hub and is described in 5.1.6. The Dashboard shows a summary card of the same data with the **"See full detail"** link.

4.3. Azure Advisor

Direct sync with Microsoft's native recommendations, classified into Cost, Security, Reliability, Performance, and Operational Excellence. Recommendations display in the language you have active on the platform (not Azure's original language).

How to use it: filter by category, review each recommendation's potential savings impact, execute remediation commands, or manage lifecycle states:

- **Snooze for 30 or 90 days:** If the optimization requires prior team validation or a future maintenance window. The recommendation disappears from active views and does not count towards pending open balances in the **Optimization Index (COIN)** or potential savings. When the time expires, the platform automatically reopens it if it remains unresolved.
- **Dismiss (Permanent):** If the recommendation does not apply to your architecture. It transitions to the *Dismissed* state indefinitely and is excluded from active views.

> ℹ️ **Scope clarification regarding Azure Portal vs. FinOps Platform:**
>
> - **Azure → Platform Sync:** If you suppress a recommendation directly in the Microsoft Azure Portal, our platform automatically detects it and hides it from the UI.
> - **Platform → Azure Governance:** When suppressing or dismissing from the FinOps platform, the action is recorded and governed in the tenant database under the principle of least privilege (without requiring write permissions on Azure). Therefore, the recommendation **may still appear in the native Azure Portal**, while in the FinOps platform it remains formally suppressed, audited, and excluded from cost savings totals.

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4.3.1. Azure AI Document Intelligence

Under `Financial Intelligence → Azure AI → Document Intelligence`, review MTD cost, processed pages, Prebuilt/Custom usage and training hours. Select MTD/30D/90D, filter the table by model, Resource Group or subscription, and open **Optimize** to evaluate Commitment Tier, F0 or Custom→Prebuilt migration. A real tenant without accounts shows an empty state and never demo data.

4.4. FinOps Maturity

Interactive assessment that places your organization at Crawl / Walk / Run across 5 pillars (visibility, optimization, governance, intelligence, operations). Answer the questionnaire to get a score per pillar plus a personalized improvement roadmap.

4.5. FinOps Academy

Interactive learning center on FinOps and Azure cost optimization — structured courses, videos, glossary, and downloadable resources.

> ⚠️ **Mandatory gate:** new users must complete the Academy before accessing the rest of the platform (with a disclaimer explaining why). Progress is tracked **per user**, not per organization — every new person on the tenant has to complete it, and once completed it can't be re-run.

4.6-4.10. Historical Progress, TOP Expenses, Resources, Green FinOps, Captured Savings, Financial Leaks (Professional/Business+)

- **Historical Progress:** monthly trend charts, savings milestones reached, month-over-month and year-over-year comparisons. You can add manual annotations about events that explain a change (e.g. "migration to reservations").
- **TOP Expenses:** ranking of your biggest spend sources by service, subscription or resource group, with drilldown to the individual resource level.
- **Resources** (Business+): full inventory — filtering and advanced search, bulk tagging, export. A child resource with no charge of its own (for example a VM extension) does not add its cost twice: it is shown in gray as *in · $amount*, and the amount counts once, on the parent.
- **Green FinOps** (Professional+): carbon-footprint estimate by service, comparison of the most efficient regions, sustainability recommendations.
- **Captured Savings** (Professional+): tracking of real savings already achieved through RIs, Savings Plans and Hybrid Benefit, with ROI calculation. The history also includes what your team saved **directly in Azure**, without going through the platform (a disk deleted from the portal, a VM shut down by CLI): it is detected because the resource's spend drops and, when the Azure Activity Log still keeps the event (90 days), it shows who did it and when.
- **Financial Leaks** (Professional+): identifies wasted money (abandoned resources, over-sizing, unnecessary redundancy) with a remediation calendar.

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5. Financial Intelligence Section

This is where the cost and optimization analysis of Azure lives. Almost every menu entry is a **hub with tabs** (3.3). The minimum plan appears next to each name; if yours does not include it, the screen says so and offers the upgrade or the standalone module from the Marketplace (8.10).

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Hub	Plan	What it is for
Consumption & Budget	Professional+	Real spend, budgets, cost groups and forecast
Team leaderboard	Enterprise	Which team improved, cost of goods sold and gross margin
Perspectives	Enterprise	Split the spend by your own dimensions
Optimization & Savings	Enterprise	Rightsizing, rates, commitments, Hybrid Benefit, zero cost
Databases	Business+	Cosmos DB, SQL, PostgreSQL, MySQL, MongoDB, Redis and Fabric
Compute	Business+	VMs, VMSS, App Service, Functions, AVD, containers, Kubernetes and OpenShift
Storage	Enterprise	Storage Accounts, disks, backups and Data Lake
Networking	Business+	Network analysis, connectivity, balancing, egress and link capacity
Users & Licenses	Enterprise	M365 licenses and user activity
Azure AI	Enterprise	Cost and consumption of Azure's AI services
Azure Integration Services	Enterprise	Logic Apps, APIM, Service Bus, Event Grid, Event Hubs and ADF
Monitoring	Enterprise	Application Insights, Log Analytics, Monitor, Sentinel and alerts
Security	Enterprise	Defender, Sentinel, Key Vault, Entra ID, WAF and DDoS
Advanced Analytics	Enterprise	Unit Economics, allocation, scorecard, anomalies, health, What-If and MACC

5.1. Consumption & Budget (`/intelligence/consumo-y-presupuesto`)

Seven tabs: **Real Consumption**, **By Category**, **Budgets**, **Cost Groups**, **Cost Group Budgets**, **Spend & Forecast** and **CSV Ingestion**.

5.1.1. Real Consumption (Professional+)

The detailed billing dashboard, live from Azure.

How to use it:

1. Filter by period, subscription and resource group.
2. Zoom into a specific service to see its breakdown by resource.
3. Download the invoice or export the data to Excel from the corresponding button.
4. Create threshold alerts from the same page (it takes you to the Self-Service Alerts form with the context preloaded).

> If a resource spends on two Azure services (a storage account on *Storage* and on *Bandwidth*, a VM on *Virtual Machines* and on *Managed Disks*), it appears **in full under its main service** instead of split. The breakdown never adds up to more than its card's total, but a resource may be attributed to a service that is not the only place where it spends.

5.1.2. By Category (Business+)

Cost distribution by FinOps category (Compute, Storage, Networking, AI, etc.), to answer "which service family is eating my bill?" without going through service by service.

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5.1.3. Budgets (Professional+)

1. **Create a budget:** name, period (monthly/quarterly/annual), limit in $.
2. **Alert threshold:** you define at what % of the budget you want to be notified (e.g. 75%).
3. The system tracks automatically — you see real consumption vs. budget in real time, with a history of previous periods.

5.1.4. Cost Groups (Business+)

Custom cost grouping based on your own business logic (by project, line of business, application or environment).

1. You create a group with naming rules (which resources go in, by name pattern or tag).
2. You assign resources manually if adjustments are needed.
3. Clicking a group opens a modal with tabs: **Costs**, **Actions**, **Resources**, **Governance** — each with the detail for that group.
4. You compare groups with each other from the list view. Among the available periods is your tenant's **Fiscal Year** (5.1.5).

This screen and its detail respect the **Perspective scope** of the top bar (3.4).

5.1.5. Cost Group Budgets (Enterprise)

You define cost centers, assign resources (manually or in bulk) and the system allocates spend automatically to generate **internal chargeback** by department, against each center's monthly budget.

Budget rollover. A budget can carry unspent money over to the next month, with a cap. The month is measured against the base budget plus what was carried over, and the card shows it with a badge and its breakdown: *"Base budget X plus Y carried over from previous months of the fiscal year"*. Rules worth knowing:

- A month in the red carries over **zero**, never a negative balance: a one-off overspend does not put the following months in debt without anyone deciding it.
- The cap is applied **at every step**, month by month.
- With rollover off —the default— the dashboard is identical to the usual one.
- It is turned on per cost center through the API: `PUT /api/intelligence/cost-centers` with `rolloverEnabled` and, optionally, `rolloverCapUsd` (Admin or Owner).

Fiscal year. The annual accumulations —the Whiteboard's, the *Fiscal Year* period of Cost Groups and the forecast— count from the month in which your **tenant's fiscal year** starts, which defaults to January. If yours starts in another month (for example April), it is configured with `PUT /api/admin/tenant-settings` (`fiscalYearStartMonth`, 1 to 12; Admin or Owner) or by asking support. The annual forecast divides by the days of the fiscal year and not by the days since January, so it does not come out low when the fiscal year is not the calendar year.

Under a Perspective scope (3.4) the budget is **not narrowed**: it is defined for the whole center, and comparing it against a slice would show a margin that does not exist. The screen says so instead of leaving the cell empty.

5.1.6. Spend & Forecast (Enterprise)

It combines two tools in one page:

- **Cost histogram:** daily distribution of spend, with a range selector from the last month up to **13 months back** (everything Azure Cost Management allows querying), cached in Redis for an instant response.
- **Spend Projection:** calculate how much you will spend in the future.
    1. The system takes the average monthly spend of the last 12 months as the base.
    2. You enter an **expected annual growth %** (you can use negative values to simulate an optimization/savings scenario).
    3. You choose the horizon: 3, 6, 12 or 24 months.

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4. The result shows: base average, equivalent monthly rate, and the projected total, with a real vs. projected line chart.

There is a summary card of the same data on the Dashboard with the **"See full detail"** link. Both cards respect the Perspective scope; the comparison against the budget does not (5.1.5).

5.1.7. CSV Ingestion (Business+)

Manual or scheduled import of cost files in the **FOCUS** standard and of Azure exports: you upload the third-party billing CSV and analyze it alongside your Azure data.

5.2. Team leaderboard (`/intelligence/leaderboard` , Enterprise)

Who improved, not who spends least. The dashboard compares the **groups of a Perspective** (3.3 and 5.3) and sorts them by improvement, which is the behavior you want to reward: a ranking by absolute spend would punish the team that runs production and reward the small one.

Column	What it shows
Team	The Perspective group
Spend	What was spent in the period
Change	Against the previous period
Budget used	How much of the budget it consumed
Captured savings	Remediations approved **and executed** in Azure for that group

How it is sorted. By the percentage drop against the previous period and, at equal change, by captured savings. Two cases stay **outside the order** but in view, with their reason: a group with *no previous period to compare against* (it neither improved nor worsened) and one *too small to rank without it being noise* (below 1% of spend: one that goes from 1 to 0.50 dollars does not beat the one that cut thousands). Each row has **See breakdown** to reach that group's detail: a ranking that cannot be explained creates arguments, not optimization.

Three things that clarify the numbers:

- **The budget is matched by name.** A Perspective group and a cost center are different things; if the group's name matches a cost center's, its budget is shown, and if not, the cell stays empty instead of showing another team's.
- **Captured savings come from what was executed**, not merely approved. Since remediations do not store tags, in a Perspective defined by tags those savings fall into *unassigned*.
- **Unassigned spend** (no rule reaches it) is part of the total but of no team, and is reported separately.

Cost of goods sold and operating expense. A card splits the spend into **COGS**, **OpEx** and **Unclassified**, and the sum matches the total. Each group is classified with `PUT /api/intelligence/leaderboard` (`perspectiveId` , `groupName` , `classification` = `COGS` , `OPEX` or `UNCLASSIFIED` ; Admin or Owner): the screen shows the result, the classification is assigned through the API.

Gross margin. When you load **revenue**, the card calculates *revenue − COGS*. It is sent with `POST /api/unit-metrics/ingest` , with a public API Key that has the `write:metrics` permission and `metricType` = `REVENUE` (11.11). Honesty rules of the calculation:

- **No revenue means no margin**, not 0%: the card explains how to send it.
- **If there is unclassified spend, the margin is a range**: next to the best case the worst case is shown (everything unclassified counted as COGS), because an underestimated COGS makes the margin look better than it is.
- A **negative margin** is shown as is: spending more than comes in is exactly what this screen exists to show.

Without Perspectives defined the dashboard says so and takes you to **Define one**. Users with an assigned scope do not see it: a ranking compares groups with each other and their slice is precisely not seeing the others.

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5.3. Perspectives (`/intelligence/perspectives` , Enterprise)

A **Perspective** splits the spend along a dimension of your business —teams, products, environments, cost centers— without touching Azure tags. Each one is an **ordered** set of rules that assign a resource to a group.

Three properties that make it auditable:

1. **A resource falls into a single group per Perspective**, and the **first matching rule** decides: order matters and is editable.
2. **The same resource can be in groups of different Perspectives.** The same disk can be "Production" in *Environments* and "Engineering" in *Cost Centers*.
3. **The sum of the groups plus *Unassigned* equals exactly the period's total.** The *Unassigned* group is always shown, with its amount and percentage: a Perspective that covers 60% of the spend and does not say so would read as if it were the total.

Creating a Perspective.

1. In **New Perspective** type the name (for example *Teams*, *Products* or *Environments*) and press **Create**. It is born **without rules**, on purpose: with an empty rule the split would show a nameless group and 100% unassigned, which reads as if something had failed.
2. Add rules with **Add rule**. Each rule has a **Group** (the name of the team, product or environment) and a criterion:

Rule type	What it compares
Tag	Key and value of a resource tag (the value accepts `*`)
Name pattern	The resource name
Subscription	The Azure subscription
Resource group	The resource group
Resource type	The Azure resource type

3. Order them with **Move up** and **Move down**: the first one that matches wins.
4. **Preview split** calculates, without saving anything, how much of the period's spend falls into each group and how much stays *Unassigned*, with the coverage percentage. Changing the order of the rules changes the split in the preview.
5. **Save** applies the rules.

Delete a Perspective asks for confirmation with its name in view: its rules are deleted too, and whoever has it assigned as scope stops having it.

"Unassigned" and "No data to evaluate" are not the same thing. *Unassigned* is the spend no rule reached: it is what you still have to tag or a rule still to write. *No data to evaluate* appears when the rules need a piece of data the platform does not yet ingest for those rows —for example, tag rules on a tenant whose cost sync does not bring tags at resource level—: it is not a problem with your tagging. Setting up the **Azure FOCUS export** completes those tags.

Where it applies. With a Perspective and a group selected in the top bar (3.4), these narrow their content: Cost Groups and their detail, Cost Group Budgets and their resource inventory, Spend & Forecast, the Whiteboard, Storage, the What-If simulator and the FOCUS export (`/api/exports/focus?perspective=&pgroup=` , useful to hand finance only one team's slice). When a screen **cannot narrow** precisely, it does not show you a dubious slice: it says so (*"Could not narrow by Perspective… the numbers below are those of the full dashboard"*).

Scope assigned to a person. An Admin can leave a user *scoped* to a group (8.2). For that person the cut is not a preference: the API applies it, so changing a parameter in the browser does not show more, and the screens that do not know how to narrow answer with access denied instead of showing the whole dashboard.

In the demo, Perspectives are examples and no changes are saved.

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5.4. Optimization & Savings (`/intelligence/optimizacion-y-ahorro` , Enterprise)

Seven tabs: **Rightsizing**, **Rate Optimization**, **Hybrid Benefits**, **Reserved Instances (RIs)**, **Saving Plan**, **Zero Cost** and **Optimization Index**.

5.4.1. Rightsizing

Analyzes 30 days of real usage and recommends the optimal SKU; shows the estimated savings before you approve the change. Dedicated verticals at `/intelligence/rightsizing/{vmss,appservice,sqldb,storage}`.

5.4.2. Rate Optimization

Rate opportunities —savings plans, Spot and volume discounts—, comparing your current rates against market benchmarks to prepare a negotiation with Microsoft.

5.4.3. Hybrid Benefits (AHB)

Shows which VMs could use licenses with Software Assurance (Windows Server and SQL Server) and how much you would save by enabling it; tracking of what already uses it.

5.4.4. Reserved Instances (RIs) — Commitment Discounts

Besides global coverage and utilization, the **Active Reservations** table replicates Azure's *Reservations* blade: Name, Status, Expiration, Scope, Type, Product, Region, Renewal, Quantity, and utilization for the last day and the last 7 days.

- Click the **Renewal** button → modal to **enable/disable auto-renewal** for that reservation. The change is applied directly in Azure — it requires the tenant's Admin/Owner role **and** `Reservations Contributor/Owner` permissions in Azure.
- Click any **utilization percentage** → modal with last day / 7 days / 30 days detail and the daily trend.
- **Orphaned reservations** —still billing without real consumption— are flagged with an alert so you can check whether the resource exists or cancel the reservation.

Above the dashboard, three readings of the portfolio:

- **Effective savings rate (ESR).** How much you really save against paying everything at list price: `(list price − effective cost) / list price`, over the last 30 days. The effective cost **includes the commitment that went unused**, so buying too much does not improve the number. It is calculated only on spend that has a list price, and the screen says on what percentage: whatever could not be valued stays out instead of being counted as zero discount. If the tenant does not yet have the cost **FOCUS export** configured, there is no list price and the ESR is not drawn.
- **Expires together.** What part of the committed value expires in the same month or in a rolling three-month window: if the whole portfolio expires at once, the spend goes back to list price all at once. It is weighted by **value**, not by count (ten small reservations do not weigh like one large one), and it says how many reservations were left without cost or without a date so as not to understate the risk.
- **Annual refund limit (US$ 50,000).** Microsoft caps reservation cancellations and refunds at **US$ 50,000 per year** per Billing Account, in a rolling 12-month window. A bar shows how much of the quota you have used and how much is left, and warns once you pass 80%.

Break-even and optimal mix simulator. Choose the workload type (**General Mix**, **Flexible Compute** or **Databases**) and your monthly pay-as-you-go (PAYG) spend: the simulator calculates in which month a **1-year** (~38% reference discount) or **3-year** (~62%) reservation beats paying on demand, how much you save per month after that point, and suggests a **3-layer mix**: Savings Plans for flexibility, reservations for stable workloads such as databases, and a pay-as-you-go cushion (*elastic PAYG*) to absorb peaks.

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5.4.5. Saving Plan

Savings Plan vs Reservation comparator: simulate both purchase options with your real numbers and compare the savings over 1 and 3 years before committing. Savings Plans for Compute are flexible hourly spend commitments.

5.4.6. Zero Cost

Inventory of zombie, orphaned and powered-off resources that generate residual costs, and which resources are free in your subscription (free tier, credits) so you can make the most of them.

5.4.7. Optimization Index (COIN)

Consolidated indicator of efficiency and use of the spend: a composite 0-100 score, with the adoption rate of recommendations, the state funnel (pending, accepted, implemented, snoozed, dismissed), the breakdown by the 5 Well-Architected Framework pillars and the historical series against the excellence benchmark.

5.5. Databases (`/intelligence/bases-de-datos` , Business+)

Visibility, metrics and cost recommendations for Azure databases. Seven tabs:

Tab	What it covers
CosmosDB	RU consumption, partitioning and costs
Azure SQL / Managed Instance	DTUs, vCores, Elastic Pools and Managed Instances
PostgreSQL	Costs, IOPS and scaling on Flexible Server
MongoDB	Mongo clusters and API compatibility in Cosmos DB for MongoDB
Azure Cache for Redis	Memory, hit rate and sizing; includes detailed monitoring of the 12 critical Azure Cache for Redis metrics
MySQL	Compute, storage and high availability on Flexible Server
Microsoft Fabric	F-SKU capacities, OneLake storage and analytical consumption

As in the rest of the cockpits, if Azure does not return a metric the platform shows **"no data"** and not a zero: a zero would read as "idle" and lead you to shrink something that may be working.

Cosmos DB: real telemetry and RU/s optimizer. Each account is measured with the real Azure Monitor metrics (normalized RU, provisioned throughput, requests and those throttled with a 429 error, storage, latency) and the dashboard issues three decisions, always with the saving derived from what was measured and from published rates:

1. **Lower the manual RU/s** when it is oversized: it is sized by the **95th percentile of usage with a 25% margin** — not by the average, which leaves the database short at every peak— and never below the 400 RU/s minimum.
2. **Move to Autoscale** when average usage is low relative to the ceiling: the threshold (~66%) comes from the ratio between the two published rates, it is not a chosen number.
3. **Switch to Serverless** when the account consumes little and intermittently: it is decided with the RUs actually consumed extrapolated to the month, and the 5,000 RU/s ceiling is measured against the **real peak**, not against what is provisioned.

It also detects **hot partitions** (one physical partition concentrating the consumption while the others have room to spare). That recommendation carries a saving of **zero** on purpose: spreading the key better prevents 429 errors, it does not lower the bill by itself, and putting a number there would be making it up.

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5.6. Compute (`/intelligence/computo`, Business+)

Efficiency and costs of the compute platforms. Nine tabs:

Tab	What it covers
Compute Efficiency (Enterprise)	Unit cost per core, regional benchmark and commitment optimization. In the **By SKU and Architecture** table, *vCPU/VM* and *RAM/VM* are the spec sheet of **one** instance of the SKU, *Instances* tells you how many you have, and the *$/Core* and *$/GiB* ratios divide the total cost by that SKU's fleet
Kubernetes (Enterprise)	AKS clusters: active nodes, real utilization vs. over-provisioning, infrastructure and control-plane cost, cost per pod, auto-scaling recommendations and **chargeback by namespace** (you assign namespaces to teams and it calculates how much each one spends, for internal billing)
Container Structure	Azure Container Apps, Registries and Environments: financial inventory, forecast and rightsizing; monthly cost per app, CPU/memory and replicas, and *scale-to-zero* opportunities (apps with a minimum replica ≥ 1 that could be switched off with no traffic)
Web Apps & App Services	Cockpit for App Service Plans and Web Apps with app density, consolidation and detection of underused instances
Function Apps	Invocations, GB-s, log auditing, dedicated and flexible plans, and sampling/downgrade playbooks
Azure Virtual Machines	Compute cost separated from persistent storage (leaks in deallocated VMs), AHUB, rightsizing to B-series, 8x5 schedules, discarding abandoned VMs and Spot and Savings Plans opportunities
Azure Virtual Desktop	See below
Virtual Machine Scale Sets	Elasticity, autoscale policies and Spot instances
Azure Red Hat OpenShift	Cost governance, node capacity and TCO of ARO clusters

Azure Virtual Desktop. Inventory of **Host Pools** and their **Session Hosts** (the underlying VM and its real cost), with KPIs for host pools, session hosts, active sessions, monthly compute cost and the cost of **FSLogix profile storage** (detected by resource name: Azure offers no direct link). Per user it shows connection hours, peak concurrent users and **cost per user** for the last days; it also lists *application groups*, *workspaces* and user assignments. A host pool whose application groups are not published to any workspace is flagged **UNREACHABLE**: nobody can connect to it even though it keeps costing. Remediation recommendations come with the **View remediation** button.

5.7. Storage (`/intelligence/almacenamiento`, Enterprise)

Visibility and cost optimization of storage services. Four tabs: **Storage Accounts**, **Managed Disk**, **Backups** and **Data Lake Gen2**.

Each is a *cockpit* with the same layout: KPIs for MTD cost, end-of-month forecast, potential savings, change against the previous month, resources evaluated, efficiency, low utilization and operational health; filters by resource, tier and resource group; sorting and prioritized recommendations.

Besides the tabs, **storage efficiency** (`/intelligence/storage-efficiency`) simulates the saving of moving blobs between Hot/Cool/Archive tiers before you apply it. With a Perspective scope active (3.4), the dashboard is narrowed with the cost that can be attributed per resource; if there is nothing to narrow with, it says so instead of showing "your group has no storage".

5.8. Networking (`/intelligence/redes`, Business+)

Network analysis, connectivity, balancing and internet access. Eight tabs:

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Tab	What it covers
Network Analysis	Breakdown by resource of all the tenant's network services: bandwidth, gateways, load balancers and public IPs; transfer peaks and idle IPs to remove
Basic Networks	Topology, private connectivity and cost observability in Virtual Networks, Private Endpoints, DNS, NSGs and route tables (UDR)
Hybrid Connectivity	ExpressRoute, Virtual WAN, VPN Gateways and connections: tunnel optimization, ExpressRoute arbitrage, idle gateways and dropped tunnels
Balancing & Publishing	Application Gateways / WAF, Azure Front Door, Load Balancers and Traffic Manager: orphaned balancers, autoscale and Front Door arbitrage
Internet Access	Public IPs, NAT Gateways, Azure Firewall and DDoS Protection: orphaned IPs, DDoS plans and firewall rightsizing
Traffic & egress	See below
Link capacity	See below
DDoS Protection	DDoS IP Protection and Network Protection plans

Traffic & egress. Where the money for moving data goes, which does not show up in the cost of the network resources. It classifies the billed transfer cost into **internet outbound**, **between availability zones**, **processed by NAT Gateway**, **between regions of the continent**, **between continents** and **inbound**, with cost and volume per region and meter, over a window of the last N days. The cost is what Azure billed, not an estimate. It gives suggestions for the two avoidable traffics: traffic **between zones** is paid in both directions and often comes from replicas or balancing spread out unnecessarily; the **NAT Gateway** charges per GB processed and, if the destination is an Azure service, a Private Endpoint or a Service Endpoint avoids it entirely. A limitation the screen states: the **volume is a floor** (rows with no charge, such as the free outbound tier, are not ingested), while the cost is the total.

Link capacity. It compares the **contracted** bandwidth of each ExpressRoute circuit and VPN gateway against the **P95 of real usage over the last 30 days**. Those that use less than **15%** of what is contracted are flagged as **oversized**: they can be moved down a SKU. The rule that governs is safety: **without metrics, no downsizing is recommended**. A link with no samples is not idle, most likely the read permission is missing or it is new, and lowering the SKU of a saturated one would cut the link; those links are counted separately and flagged above the table. The P95 is calculated over hourly samples (Azure Monitor offers no percentiles), and the screen shows the review date of the SKU catalog.

5.9. Users & Licenses (`/intelligence/licenses` , Enterprise)

Three tabs: **Dashboard** (M365 and Entra ID KPIs), **User Activity** (filterable table) and **License Optimization** (resources without Hybrid Benefit via Resource Graph and per-SKU metrics via Microsoft Graph).

5.10. Azure AI (`/intelligence/azure-ai` , Enterprise)

Azure's artificial-intelligence platform: vector search, document processing, transcription, vision, moderation, MLOps and distributed analytics. Nine tabs: **Overview**, **Azure Foundry**, **AI Search**, **Document Intelligence**, **Speech & Language**, **Vision & Video**, **Content Safety**, **Machine Learning** and **Databricks**.

The **Azure Foundry** tab is the *AI Cost Analytics*: cost per model and token consumption in Microsoft Foundry / Azure OpenAI, with call-optimization recommendations. For Document Intelligence see 4.3.1.

5.11. Azure Integration Services (`/intelligence/integration-services` , Enterprise)

Operation and cost of enterprise integrations. Six tabs: **Azure Logic Apps** (with the distinction between Standard and Enterprise connectors and its cost impact), **Azure API Management (APIM)**, **Azure Service Bus**, **Azure Event Grid**, **Azure Event Hubs** and **Azure Data Factory (ADF)**.

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5.12. Monitoring (`/intelligence/monitoreo` , Enterprise)

Observability and operation of the monitoring services. Eight tabs: **Application Insights**, **Log Analytics Workspace**, **Azure Monitor**, **Microsoft Sentinel**, **Alerts**, **Action Groups**, **Workbooks** and **Network Watcher**.

In **Log Analytics Workspace** you control monthly cost, retention and estimated ingestion per workspace; it detects unnecessary massive ingestion (workspaces without a daily cap), excessive retention and *Commitment Tier* opportunities, with estimated potential savings. It also shows as a card on the White Board.

5.13. Security (`/intelligence/seguridad` , Enterprise)

Visibility and cost control of the security services. Six tabs: **Defender for Cloud** (per-resource protection plans: Servers, Containers, SQL, Storage), **Microsoft Sentinel** (log ingestion and capacity commitments), **Key Vault**, **Entra ID** (P1/P2 licensing and workload identities), **WAF** and **DDoS Protection**. A pre-warm cron speeds up loading.

5.14. Advanced Analytics (`/intelligence/analitica-avanzada` , Enterprise)

Advanced models for allocation, anomalies, health and financial simulation. Seven tabs:

Tab	What it does
Unit Economics	You define your own unit metric (cost per transaction, per user, per MB processed) and the system calculates the unit cost over your Azure data: total spend of the period divided by total volume, not the average of the daily averages. The business volume is loaded by hand in the screen's configuration or automated from your pipeline with the ingestion endpoint (11.11)
Allocation	Rules to distribute shared costs across areas, cost centers and teams (by real usage, proportional or fixed)
Scorecard (Fama)	0-100 score of your financial health with individual indicators (savings, reservation coverage, efficiency) and a maturity assessment in *Inform, Optimize and Operate* (FAMA framework); you can set goals
Anomaly Detection	Identifies abnormal spend spikes or drops (Z-Score) and lets you create alerts by adjusting sensitivity. Each anomaly has an owner and follow-up (18)
Tenant Health	Comprehensive index of architectural and governance health: detected problems, billing consistency, reservation coverage and policy compliance
What-If Simulator	See 5.14.1
MACC Tracking	Tracking of the minimum annual commitment (EA/MCA) — consumed vs. committed, burn-rate and compliance projection

5.14.1. What-If Simulator

Simulate the impact of scaling compute/storage, varying network traffic or enabling Azure Hybrid Benefit on your current cost, **before applying the real change**.

Full flow:

1. You move the sliders: compute scale, storage scale, % increase in network traffic, enable/disable AHB.
2. You press **"Run Simulation"** — you see the breakdown: base cost, projected cost, delta and % change.
3. **Save the scenario: "Save current"** button → you give it a name and optional notes. The scenario is frozen with those numbers (it is not recalculated later, even if your real base cost changes over time — so you can compare scenarios saved at different moments with each other consistently).
4. You repeat the process 2-3 times with different sliders to have several saved scenarios.
5. **Compare:** you tick 2 to 4 scenarios with the checkboxes → **"Compare"** button → a side-by-side modal opens with each scenario as a card, marking which one is the baseline.

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6. **Export:** you choose the format (CSV, PDF or Markdown) next to the download buttons — you can export a single scenario, all saved ones, or the full comparison with each one's delta against the baseline.
7. **Delete:** trash icon on each row (only the scenario's creator or an Admin/Owner can delete it).

> The assumed calculation mix is Azure-first: 60% of the cost is compute, 25% storage, 15% network; AHB applies a flat 18% discount on the total if enabled.

With a Perspective scope active, the base spend of the last 30 days is that of the slice: "what happens if my team grows 20%?" is answered over the team's spend and not the tenant's. The list of saved scenarios remains the tenant's.

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6. Cloud Cleanup Section

6.1. Zombie Resources (`/cleanup/zombies` , Professional+; remediation Business+)

Detects orphaned and underused resources across 25 types of Azure infrastructure (unattached disks, orphaned public IPs, NICs without a VM, empty App Service Plans, Elastic Pools without databases, idle VMSS, old snapshots and deallocated VMs with active storage).

Capabilities and flow:

1. **Omni-Scan of 25 Resource Types:** Exhaustive analysis in Azure Resource Graph.
2. **Hard vs Soft Waste classification:** Visual distinction between direct monetary waste (Hard Waste) and governance/tagging misalignment (Soft Waste).
3. **Subscription resolution:** Always shows the friendly subscription name (e.g. `CSCS-LandingZone`).
4. **Customizable table:** Columns with interactive width handles (`col-resize`) and a `Customize Columns` dropdown (`IconColumns`) persisted per browser.
5. **Safe remediation:** Creation of preventive snapshots and execution of CLI/PowerShell playbooks.

6.2. Networking Zombies (`/cleanup/zombies/networking` , Professional+; remediation Business+)

Specialized audit of network infrastructure:

- **Idle VPN / ExpressRoute Gateways:** Detection of gateways with no active connections or configured IPsec tunnels.
- **Orphaned Public IPs:** Identification of IPs assigned without association to NICs or load balancers (excluding the system's Network Watchers).
- **Disconnected Private Endpoints:** Diagnosis of private endpoints in `Disconnected` state or pointing to deleted resources.
- **NAT Gateways and Firewalls:** Detection of NAT Gateways without subnets and Firewalls/Application Gateways without routing rules or backend pools.

6.3. TTL Expirations — Time-To-Live (`/cleanup/ttl` , Business+)

Automated lifecycle control of ephemeral resources (test environments, development labs and sandboxes):

1. **Centralized TTL Policies:** Definition of maximum lifetime per resource type and environment.
2. **`ExpireOn` tagging:** Optimistic application of expiration tags with date validation.
3. **Expiry traffic lights:** Real-time indicators (`CRITICAL` expired, `WARNING` expiring in less than 72h, `ACTIVE` current).
4. **Interactive extensions:** Quick lifetime extension (+7d, +14d, +30d) for resources that need more test time.

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5. **Immutable audit:** Unalterable history of the deprovisioning actions executed in `TtlDeletions`.

6.4. Orphaned Backups (`/cleanup/backup-orphans`, Professional+; remediation Business+)

Management of storage, accrued costs and legal compliance of backups in **Recovery Services Vaults**:

1. **Detection of unlinked Protected Items:** Identification of protected instances whose original resource has already been deleted from Azure but which keep accruing storage fees and base costs.
2. **Compound cost calculation:** Exact monthly cost breakdown adding the base fee per item type (`AzureIaasVM`, `AzureWorkload`, `AzureStorage`, `AzureDisk`) and the vault storage consumption at $0.0224/GB/month.
3. **Safe purge with Soft Delete:** Strict confirmation modal with name verification and a warning about the preventive 14-day Soft Delete window before permanent deletion.
4. **Legal compliance exemptions:** Side drawer to register audit tickets (SOX, tax, regulatory) with retentions from 1 to 10 years or indefinite, excluding the item from active waste.
5. **Transfer to Archive Tier:** Move to the cold storage tier to obtain up to an 85% reduction in long-term retention cost.

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7. Governance Section

7.1. Tag Governance — Azure Tag Governance Engine (`/governance/tags`, Professional+)

Comprehensive audit, AI suggestion, and metadata propagation panel in Azure.

Key capabilities:

- **Dual Compliance Audit:** Dedicated views to audit both **Individual Resources** and **Resource Groups (RGs)**.
- **4 Mandatory Policies:** Real-time compliance tracking for the 4 core FinOps tags:
  - `Environment` (prod, staging, dev, qa, test, sandbox)
  - `Role` (architectural or operational role)
  - `CostCenter` (accounting cost centre)
  - `Department` (owning business unit)
- **AI-Powered Tag Suggestions (1-click):** Engine analyzing resource type, name conventions, and subscription context to draft recommended missing tags instantly.
- **Bulk RG Tag Inheritance (Safe Merge):** Propagate tags from the parent Resource Group to child resources without overwriting or deleting pre-existing tags.
- **Optimistic Editing & Local Cache:** In-line editing with immediate persistence in local cache and asynchronous background sync to Azure Resource Manager.
- **CMP Table Standard & Customization:** Resizable column headers (`col-resize`, 100px - 600px), column visibility popover in `z-[100]`, and automatic tenant persistence in `localStorage`.

7.2. Governance Reporting (`/governance/reporting`, Enterprise+)

Unified executive dashboard. At its centre sits the **Financial Security Score**, a 0-100 index weighing four pillars: Azure Policy compliance (40%), mandatory tag hygiene (30%), RBAC assignment hygiene (20%) and zombie resource control (10%).

When a pillar cannot be measured —because the Service Principal lacks permissions, or because no policies are assigned— it counts as neither 0 nor 100: it is flagged "not measurable" and its weight is redistributed across the rest. "View pillar breakdown" shows each pillar's nominal and effective weight.

What else you get:

- Azure Policy compliance with an expandable detail of non-compliant resources.

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- Resource inventory by type and by region, with everything outside the top grouped under "Other" so the bars add up to the headline total.
- RBAC assignments by principal type, including an **orphaned SID audit**: assignments whose user or application no longer exists in the directory. They grant nobody access today, but if Azure reuses that identifier the permission comes back to life on a different principal.
- Export to **executive PDF** and to **CSV** with the full dataset.

7.3. Virtual Machine Control — Power Schedules (`/governance/power` , Business+)

Automated VM start/stop routines outside business hours, plus live manual control.

Two scheduling modes:

- **One-time:** runs the action (start / stop / restart) once, at an exact date and time.
- **Recurring:** you pick the time and the weekdays. A "from–to" window is built from two rules: one to start and one to stop, over the same days.

Smart Shutdown. Before every scheduled shutdown the system checks the VM's actual CPU over the last 30 minutes. If it sits above the threshold (5% by default, adjustable), it **postpones the shutdown** and records it as skipped rather than taking down a machine that is working. Tune the threshold with "Calibrate Threshold": too high shuts down machines mid-job, too low lets OS background noise cancel every shutdown and the savings never materialise.

Off-hours savings. The first KPI shows the monthly spend you already recover through active schedules, and how much more is available on running VMs with no schedule. The calculation uses each VM's real window —from its shutdown to its next start— rather than a constant: a weekend with no scheduled start extends Friday's shutdown all the way to Monday.

Live control. The lower table lists every VM with its state, size, current CPU and spend. Select several and apply Stop / Start / Restart in bulk. The state badge changes immediately while Azure processes the operation, and corrects itself if something fails or if Smart Shutdown skipped the shutdown.

Operational details:

- The time zone is stored as an IANA name, so a schedule keeps firing at the correct local time across daylight-saving changes.
- The system checks pending schedules every few minutes, plus an immediate check on save.
- If Azure Monitor returns no metrics for a VM, the CPU column shows **"n/a"** instead of 0%: a zero would read as "idle" and could lead you to shut down a machine with no telemetry.

7.4. High Availability Recommendations (`/governance/ha` , Business+)

Detects virtual machines, databases and cloud resources without zonal or geographic redundancy, or without backup.

Every gap tells you the SLA you have today and the one you would reach, translated into minutes of monthly downtime — because "99.9%" means nothing until you turn it into 43.2 minutes a month, and 99.99% into 4.3. The "View Architecture" drawer shows the topology, the SLA comparison and the estimated additional cost before you decide.

What is detected: VMs with no zone or Availability Set, resources with no backup, databases without a failover group or geo-redundancy, single-instance App Service Plans, and Basic SKU public IPs.

About Basic IPs: they show an SLA of 0, not 99.9. Microsoft publishes no SLA for that SKU — it is not that it is low, it is that there is none.

About backups: a backup does not improve availability, it improves RPO (how much data you lose if something breaks). That is why current and projected SLA match on those rows: we do not promise an improvement that does not happen.

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What can be remediated from here: only the public IP SKU upgrade and attaching a backup policy, both idempotent Azure operations. Spreading across zones or enabling geo-redundancy requires recreating the resource or choosing a secondary region — for those, "Remediate" hands you the change blueprint with the exact command so your infrastructure team can plan it.

Exemptions. If a non-production workload does not warrant redundancy, exempt it with a justification. It stops counting towards the KPIs but stays visible under the history button, and the justification is recorded against your user — somebody will have to defend it at the next audit.

7.5. Expiring Credentials — Entra ID (`/governance/credentials` , Business+)

Inventory of App Registration and Service Principal secrets and certificates, across two tabs.

Credentials tab. Each row shows the status —**Healthy** (more than 30 days), **Expiring Soon** (30 days or fewer) or **Expired**—, the expiry date, the exact days remaining and the App ID with a copy button. Days are recalculated on every query: a credential moves from healthy to expiring without anyone touching it.

Secret rotation. The "Rotate" button generates a new secret through Microsoft Graph with the validity you choose (6, 12 or 24 months).

> **Important:** rotation **does not revoke the previous secret**, and that is deliberate. Revoking in the same step would take down everything still using it — precisely the incident this module prevents. The correct order is: rotate, migrate the consumers to the new secret, and only then delete the old one from the Entra ID portal.
>
> The secret value **is shown only once**. Microsoft Graph never returns it again and the platform stores it nowhere. Copy it there and then, and keep it in Azure Key Vault.

Certificates are not rotated here: they are renewed by uploading the public key, a different procedure.

Configured Alerts tab. Advance-warning rules. Each rule accepts **several thresholds** (60, 30 and 7 days, for example), and each one fires an independent notice — a single reminder at 7 days rarely leaves enough time to coordinate a rotation across every consuming team. Pick the channels (Email, Teams, Slack, Webhook) and the recipients, and enable or disable each rule with a switch.

7.6. Auto-Block Policies (`/governance/policies` , Enterprise+)

Cost prevention at provisioning time through Azure Policy: constraints that stop unwanted spend from coming into existence.

What you see: your environment's overall compliance, the breakdown by resource category, the status of each governance initiative, and the table of active policies with their effect (`Deny` , `Modify` , `Audit` , `DeployIfNotExists`), their scope and how many resources each one is breaching.

Everything comes from live Azure Policy. **If your environment has no policies assigned you will see 0 evaluations** and a message explaining why — not an estimated percentage.

Deploying a policy. The wizard lets you pick the scope (management group or subscription) and one of the predefined templates: restrict VM sizes, block public IPs in sandbox, inherit tags, require CostCenter, restrict regions, or audit storage without HTTPS. Each template explains the cost it prevents.

> A `Deny` policy **blocks new deployments but does not revert what already exists**. Resources created before the assignment will show as non-compliant until you fix them by hand or with a `Modify` policy. Azure takes up to 30 minutes to complete the first evaluation.

Remediate. The remediation button creates an Azure Policy task that fixes existing resources. It only appears on `Modify` and `DeployIfNotExists` policies: those are the only effects Azure can apply retroactively. It is not offered on `Deny` or `Audit` , where the task would finish with zero resources fixed.

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View non-compliant resources opens a side panel with each breach and its reason.

7.7. Remediation Approvals (`/governance/approvals`, Business+)

Control flow over the infrastructure changes proposed by the optimisation engines. It runs on the **four-eyes principle**: whoever requests a change cannot approve it.

Pending requests. Each card shows the resource, the proposed action, who requested it, the monthly savings it releases and any relevant warnings — whether the VM will reboot, whether the action is irreversible.

Approving executes the change in Azure immediately. It is not a status change in a list: the platform calls Azure Resource Manager and the real outcome lands in the history. If Azure rejects the operation, the request shows as **Failed** with the literal error, not as approved.

Safety controls:

- **Pre-deletion snapshot.** When deleting a disk you can request a backup snapshot. If the snapshot fails the deletion **does not run**: you asked for a safety net, and without it the action does not proceed.
- **"Approve everything safe"** only covers non-destructive actions that do not restart a service. Deleting a disk or resizing a production VM requires a conscious decision, not a bulk click.
- **Rejecting requires a reason**, which is sent to the requester. Without one, the same request comes back next week.

Decision history. Full audit trail: who resolved it, when, what Azure replied and —if one was created— the backup snapshot identifier. The "Released Savings" KPI counts **only what Azure confirmed**: an approval that failed released nothing and is not counted.

7.8. Self-Service Alerts (`/governance/alerts`, Professional+)

Financial alert rules you build yourself, without asking support, delivered to Microsoft Teams, Slack, webhooks, email and ServiceNow. They are created with a **3-step** wizard (*Trigger and destination configuration*).

Four alert types:

Type	It fires when...
Budget (% of consumption)	spend exceeds a percentage of the budget
Fixed threshold ($ USD)	spend exceeds an amount in dollars
Statistical anomaly (+% deviation)	spend deviates from the usual by more than the chosen percentage
Month-end forecast (> % EOM)	the end-of-month projection exceeds the chosen percentage

Scope. Each rule can watch the **whole tenant**, a **subscription**, a **resource group** or a **cost center** (by tag). The **condition preview** shows in one sentence when the alert will be emitted (*"when spend on X exceeds Y under the criterion of Z"*) before you save it.

Rules table. You search by name, subscription or channel; filter by type, channel and status (active or paused); sort by most triggered, last activation, name or threshold. Each rule is **paused with a toggle**, **edited** without being duplicated, and has **"Test dispatch"**, which sends a test notification to the channel so you can verify delivery before relying on it. The *Last activation* and *Triggers* columns tell you whether the rule is alive.

Delivery channels are configured in **Configuration → Notifications** (8.5).

7.9. End of support (`/governance/lifecycle`, Business+)

What runs on a version Azure is retiring, when, and what it costs today. Staying on a retired version has two endings and both are paid for: an unbudgeted emergency migration, or paid extended support. Seeing it months ahead lets you plan it.

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It crosses your tenant's inventory against Azure's retirement calendar. Today it covers **AKS** (Kubernetes version), **PostgreSQL Flexible Server** and **MySQL Flexible Server**.

At the top, a summary: *"N resources at risk — $X per month running on something Azure retires within the year, or has already retired"*.

The table shows per resource the product, the version, the **status**, the **end of support** date (*N days left* or *expired N days ago*), the paid extended-support date when it exists, and the **monthly cost**. The status is a traffic light by window:

Status	Meaning
Retired	Azure has already retired that version
Less than 90 days	Critical: retirement is imminent
Less than a year	Upcoming: there is time to plan
Supported	Supported beyond a year
No data	Nothing can be asserted

The rule that gives the screen its name: absence of data is never "supported". A resource whose version could not be read, a version later than the last one the catalog knows, or a resource type the catalog does not cover, are marked **No data**. The same goes for cost: a resource without attributable cost shows as *no data* and is reported separately —it is not added as zero—, because adding it would lower the total and make the risk look smaller.

Where the dates come from. From Microsoft's documentation, kept in our own catalog —not from an Azure API—. That is why the screen shows the **date of the catalog's last review**: if it got stale, it says so. When Azure publishes the end of a version as a month (for example "Mar 2026") it means the **last day** of that month, and that is how it is loaded.

What it does not cover yet: Azure SQL logical servers (their version is the server's, which never retires; SQL's end of support lives in each database's compatibility level), App Service runtimes, VM series and cloud licenses.

This screen is not shown to users with an assigned scope: it is the inventory of the whole platform, not one team's slice.

7.10. Shift-Left FinOps (`/governance/shift-left` , Business+)

Cost is decided when someone writes the `.tf` , not when the invoice arrives. Shift-Left brings the economic impact of an infrastructure change into the **Pull Request** review, which is where it is still cheap to correct. It works on Terraform.

How it works. In the PR pipeline the plan is generated (`terraform plan`), a GitHub action sends it to the platform, and the platform estimates the **monthly cost delta** with **Azure list prices**. The result comes back as a comment on the Pull Request (edited on every push, not accumulated) and can **fail the check** if the increase exceeds a threshold. The action and its workflow example are described in 11.10, and it needs an API Key with the `write:shift-left` permission.

How to read the number. The delta is a **floor, not a total**: what the platform knows how to map to a meter of the Azure Retail Prices API is priced —today **Linux and Windows virtual machines, managed disks and public IPs**—; the rest is counted apart as *unpriced* and appears in the comment, and if that number is not zero the real cost is higher. They are **list prices**: they do not include your reservations, Savings Plans, EA/CSP discounts or markup. It is useful to compare the before and after of a PR; it does not replace the real cost the platform shows.

What you see on the screen.

- Four KPIs: **Plans analyzed**, **Increase detected**, **Savings detected** and **Blocked by threshold**.
- The **history**: one row per `terraform plan` analyzed, with repository, author, cost *before* and *after*, **monthly delta**, threshold status (*Within threshold*, *Threshold exceeded* or *No threshold*) and date. If some resources could not be priced, the screen warns that **the delta is a floor, not the total**.

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Budget Gates. You define, per repository, how many USD per month a Pull Request may add. If it exceeds that, the GitHub check fails. How the thresholds combine:

- The workflow's `threshold-usd`, if present, **wins** over the platform's: a team can be stricter in its own pipeline, not looser.
- With no threshold the gate stays *not evaluated*: it reports and does not block.
- If the platform does not respond, it **does not block either**: it is a cost control, and halting a team's work because of an outage on our side is worse than letting an expensive change through.
- Only an **Admin or Owner** can change the thresholds.

Current limits. Terraform only (not Bicep) and GitHub only (not Azure DevOps). The threshold is in absolute USD — not a percentage—, and unblocking a blocked PR is resolved with the repository's own mechanisms (another commit, or branch protection rules): the platform has no approval flow of its own.

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8. Administration Section

8.1. Support (`/support` , all plans from Professional)

Any tenant user can open tickets to CSCloudSolutions and follow the conversation within the platform.

How to create a ticket:

1. Fill in subject, category (Technical / Billing / Inquiry / Feature Request), priority, and initial message.
2. You can attach screenshots or files (`jpg` , `jpeg` , `png` , `txt` , `json` ; max 5 MB per file, 10 per ticket) — kept for 60 days and then auto-deleted.
3. Support team replies appear marked with 🛟 in the thread. You can reply while it's open, and close/reopen it yourself.
4. **Quick access:** the lifebuoy icon next to your username in the header opens Support from any page. When your ticket gets a reply, you'll see a notification in the bell 🔔.

Quotas by plan:

Plan	Tickets/month	First-response SLA
Professional	20	24 h
Business	Unlimited	8 h
Enterprise	Unlimited	4 h

What you see on screen:

- **Four indicators at the top:** open tickets (including those waiting on your reply), in progress, resolved, and your plan's first-response SLA.
- **Troubleshooting:** a collapsible section with FAQs and a search box. Many questions are answered here without opening a ticket.
- **Your ticket table** with number, subject, category, priority, status, last update and **SLA remaining** as a live countdown. Drag the right edge of any header to resize the column, pick which columns to show with *Customize columns*, and page 15 at a time (or 30/45/60). Those preferences are saved in your browser.
- **Conversation in a side panel:** *View conversation* opens the thread on the right without losing the table. You can attach several files at once by dragging them; images show an expandable preview in the thread and logs download.

About SLA remaining: the counter measures the time until the team's **first response**. Once we reply it stops, even if the ticket stays open. If the ticket is waiting on your reply it does not run either — the ball is on your side.

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Affected module (optional): when creating the ticket you can state which module it concerns (Defender, Zombie resources, Cost allocation, Alerts, Power schedules, Tag governance). It is not required, but it speeds up routing to the right team.

Global support queue (`/superadmin/support`) — CSCloudSolutions team only

Every ticket from every tenant in one queue, with four indicators: total, **unassigned**, **SLA at risk**, and mean resolution time.

SLA at risk counts only tickets **without a first response** that have less than an hour left. A ticket that was already answered, or one waiting on the customer, does not show up here even if the clock keeps running: the ball is not with the team. The applied SLA is the one for the **owning tenant's tier**, not a single value — Professional is 24 h and Enterprise is 4 h, and the queue mixes them.

Mean resolution time averages only tickets already resolved. Including open ones would produce a number that drops when new work arrives, which is the opposite of what it measures.

Filters: status pills with global counters (not scoped to the active filter), organization, priority, and assignment (*My tickets* / *Unassigned*). The table carries number, tenant, subject, category, priority, assignee, status, SLA remaining and creation date, with resizable columns and 15/30/45/60 paging.

Per-row actions: *Attend ticket* opens the conversation; *Take* assigns it to you (and moves it to "In progress" if it was open); *Resolve* closes it as resolved. Taking a ticket **always assigns to the caller** — the client cannot assign another agent by email. **Priority** can also be changed from the queue (row selector) or from the conversation panel, without reopening the ticket.

Private internal notes. In the conversation panel, the *Private internal note* toggle leaves context for the team (diagnosis, steps taken, who to escalate to) that **the customer never sees**. The filter is server-side, not on screen: the note does not leave the backend toward a tenant user. An internal note also does not move the ticket status or count as a first response — the customer saw nothing, and the SLA reflects that.

8.2. Users & Access (`/admin/access`)

The hub has five tabs: **Users & Permissions** (what this section describes), **Security (2FA)** (section 10), **SAML SSO** (11.1), **Client Onboarding** (1.2 and 1.3) and **Lighthouse Onboarding** (Enterprise, to delegate a client's access without asking for credentials).

See section 2 for details on role vs. permissions. From the **Users & Permissions** tab you add users, edit their role, adjust their module scope, or remove their access. Only tenant Admins and Owners can open this screen.

Four indicators at the top: registered users (and how many your plan allows), admins and owners, readers, and your team's **2FA compliance**.

Adding a user — no more copying GUIDs. Type the person's name or email: the platform searches your Entra ID directory and shows the matches. Pick one and the *Entra ID (OID)* field fills in by itself, marked with a green check. Then choose the base role and the page scope (*Full module access*, *Visibility & FinOps only*, or *Cleanup & Governance only*).

Suggestions flag people who are **already added** to the tenant and accounts that are **disabled** in Entra ID.

Sync from Entra ID opens a panel with two ways to onboard in bulk:

1. **By users:** the directory listing, with multi-select and a role per person.
2. **By security group:** pick a group (for example `FinOps-Engineers`) and a default role, and all its members are provisioned at once. The **Owner role cannot be assigned by group**: ownership transfer is done user by user, deliberately. If the group has more members than your plan allows, the excess is skipped and reported.

The user table shows the name with initials, email, the abbreviated OID with a copy button, the role (editable in one click from the cell), page scope, 2FA status and last sign-in. Like every other table: resizable columns, a visible-columns selector, 15/30/45/60 paging, and preferences saved in your browser.

2FA status — three values, not two:

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Badge	What it means
2FA active	The person has a second factor registered in Entra ID.
Pending	Entra ID confirmed they have **no** second factor registered.
No data	Entra ID has not answered for that user yet. It does **not** mean 2FA is missing.

The compliance indicator is computed only over users whose status is known, and it tells you how many have no data. If the platform lacks the Graph permission to read the authentication-methods report, the column shows "No data" for everyone — not 0% compliance. The *Refresh 2FA* button queries Entra ID again whenever you need it.

Granular permissions (the *Permissions* button on each row) opens a side panel with one switch per module: Visibility, Financial intelligence, Cloud cleanup, Governance, Security, and Administration & support. Turning a module off removes it from that user's menu **and** blocks access if they try to reach it by typing the URL. Below that you can restrict which Azure subscriptions they can see: with none selected, they see every subscription in the tenant.

Team scope: restricting a person to their slice of the spend (Enterprise)

Domain permissions (section 2) decide which *pages* a person sees; **team scope** decides which *slice of the data* they see. It lets a team's owner enter the platform and see only the spend of their group of a Perspective (5.3), without seeing the rest of the organization's.

It is not a screen filter: the server applies it. With a scope assigned, the API returns only their slice —changing a URL parameter does not show more— and the dashboards' totals and percentages are calculated over that slice. Screens that do not yet know how to narrow **deny access** instead of showing the whole tenant: when in doubt, the system fails closed.

How it is assigned. Today it is through the API, done by an Admin or Owner, and only on an Enterprise tenant:

- `POST /api/admin/user-scopes` with `tenantId`, `email`, `perspectiveId` and `group` (optional: without a group, the scope is the whole Perspective).
- `GET /api/admin/user-scopes?tenantId=…` lists the current scopes.
- `DELETE /api/admin/user-scopes?tenantId=…&email=…` removes it.

The assignment response includes **how many screens are left** for that person (`screensWhenScoped`): assigning blindly would be assigning without knowing what is being taken away.

Rules.

- **One scope per user.** Assigning another replaces the previous one.
- **Reader and Colaborador only.** An **Admin or Owner cannot have a scope**: they could issue themselves an API key —which belongs to the tenant, not to them— and bypass the restriction.
- If the Perspective is deleted, the scope of whoever had it assigned is deleted with it.

What the scoped person sees. Only the screens that today know how to narrow to a group, all cut down to their slice: the Whiteboard, Cost Groups (`/intelligence/cost-groups`), Cost Group Budgets (`/intelligence/cost-centers`), Spend & Forecast (`/intelligence/cost-projection`), the Scorecard (`/intelligence/scorecard`) and the *Storage Accounts* tab of Storage (`/intelligence/almacenamiento/storage-accounts-finops-cmp`). Everything else answers with access denied. The side menu shows only the entries that are, by themselves, one of those screens (today, the Whiteboard): the others are reached by their address. The Perspective selector stays fixed with a lock and Views are not shown. It is a small product on purpose: it grows as more dashboards learn to narrow.

8.3. Configuration (`/admin/config`)

A hub with five tabs:

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Tab	What you do there
General	Tenant profile administration: name, logo, default language for new users, timezone for reports and billing cycle
AI	Enable or disable AI features, choose the model, adjust the sensitivity of detections and what data is shared (Professional+)
Notifications	Delivery channels: Slack, Teams, email and webhook (8.5)
Account Status	Your Azure subscriptions, the state of the ingestion and credential expiry; from here you unlink and relink subscriptions to choose which ones are monitored (19)
Partner Markup (CSP)	Configurable margins per service for CSP partners, applied automatically in billing (Enterprise)

8.4. Billing — Plan Change (`/admin/billing`, Professional+, Owner role)

Plan Limits:

Plan	Allowed Azure Subscriptions	Users per Tenant	Support / SLA
Professional	Up to 2 subscriptions	Up to 3 users	20 tickets/mo (24 h)
Business	Up to 3 subscriptions	Up to 5 users	Unlimited tickets (8 h response)
Enterprise	Unlimited	Unlimited	Dedicated support, unlimited tickets (4 h max SLA)

Change procedure:

1. Choose the new plan (Professional / Business / Enterprise).
2. Choose frequency (monthly/annual) and proration mode.
3. The system shows a **preview summary** with the real amount calculated by the payment gateway before confirming: *"You'll be charged $X now"* (upgrade) or *"You'll receive a $X credit"* (downgrade), the new recurring total, and the next billing date.
4. The change **only applies** when you click **Confirm change** — until then you can cancel at no cost.

8.4.1. Azure Subscription Quota Enforcement & Upgrade Modal

The platform automatically verifies connected Azure subscriptions:

- **Professional:** Up to 2 Azure subscriptions.
- **Business:** Up to 3 Azure subscriptions.
- **Enterprise:** Unlimited subscriptions with multi-account support.

If you attempt to link a subscription exceeding your quota, the **Dynamic Upgrade Modal** opens automatically to seamlessly upgrade your plan via Paddle.

8.4.2. Expanding capacity without changing plans (add-ons)

If you need more Azure subscriptions or more tenants but don't want to jump to a higher plan, go to **Administration → Billing** and look for the **Expand capacity** card.

At the top you'll see your current usage: *"You are using 3 of 3 Azure subscriptions"*.

How to purchase:

1. Enter the **total** quantity you want (not how many to add). If you already have 2 and want 5, enter `5`.
2. Press **Purchase**. If you enter a lower number, the button reads **Reduce**.
3. The change is automatically prorated on your next invoice.

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Capacity is not enabled instantly: it activates a few seconds later, once the payment gateway confirms. If you don't see it right away, wait and refresh.

List price per unit (the card shows the current price, the one the payment gateway charges):

Add-on	Professional	Business
Additional Azure subscription	$40/mo	$40/mo
Additional tenant	$90/mo	$240/mo

Enterprise does not use this card: your capacity is adjusted contractually. Contact your account executive.

If the card shows *"Online purchase is not available"*, your account has no active paid subscription (for example, you're on a trial or your plan is managed commercially). Reach out and we'll sort it out.

8.5. Notification Configuration (`/admin/notifications`, Professional+)

Three supported channels: **Slack**, **Microsoft Teams**, **Email (SMTP)**.

8.5.1. Global Notification Center (Navbar Bell)

- **Interactive Bell:** `IconBell` icon with corporate blue badge (`#0078D4`) indicating unread notifications.
- **Dynamic Filters:** Filter by *All*, *Unread*, *Info*, *Warning*, and *Critical*.
- **Quick Actions:** *"Mark all as read"* and direct deep-links to associated anomalies, reports, or support tickets.

Configure Slack:

1. Go to `https://api.slack.com/apps` → create or select an app → **Incoming Webhooks**.
2. **"Add New Webhook to Workspace"** → choose the destination channel.
3. Copy the webhook URL (starts with `https://hooks.slack.com/services/...`).
4. In the platform: **Notifications** → **Add Channel** → **Slack** → paste the URL → save.

Configure Microsoft Teams:

1. In the Teams channel → ⋯ → **Connectors** → search for "Incoming Webhook".
2. Configure the webhook and give it a name.
3. Copy the generated URL.
4. In the platform: **Notifications** → **Add Channel** → **Microsoft Teams** → paste the URL → save.

Configure Email:

- Uses SMTP. If your organization doesn't specify its own server, the platform's default SMTP configuration is used.
- You can define multiple recipients and, optionally, override host/port/user/password per channel.

Severity filter: each channel can filter what it receives — `info`, `warning`, `error`, or combinations (e.g., `warning,error` to skip informational messages).

Test the channel: a test button on each configured channel — sends a test notification before you rely on it in production.

Tier availability: Professional allows 2+ channels; Enterprise allows unlimited channels + per-channel SMTP overrides.

If notifications aren't arriving, check in this order: channel enabled (`enabled = true`) → severity filter matches the alert level → you tested the channel from the UI → there's at least one channel configured for the tenant. The send log (with errors) is available for diagnosis.

8.6. Reports & Export (`/admin/reports`, Professional+)

A hub with seven tabs:

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Tab	Plan	What it does
Executive Report	Business+	Automated generation of high-level reports, designed to present to management: a downloadable PDF executive summary, also sent by email
Report History	Business+	The executive reports already generated, to reopen or download them
Invoicing Report	Business+	Showback/chargeback invoices per client or cost center (11.4)
Workbooks	Enterprise	Custom reports: own layout, charts, data tables, exportable to PDF and shareable by link
Power BI Templates	Enterprise	Downloadable `.pbit` templates, pre-connected to your SaaS data
FOCUS 1.1 Export	Professional+	Your data in the standard FOCUS format, schedulable for automatic daily generation (11.5)
Subscriptions	Professional+	Reports by email at the cadence you choose (below)

Report subscriptions

In a FinOps tool, the report you have to go and look for does not get looked at. A subscription makes the report **arrive on its own**, at the cadence each person chooses, with a **link to the report, a chart and a CSV attachment** of the raw data (the CSV is what lets finance cross it with its own model).

Creating a subscription (Admin or Owner, because it sends email to third parties): **New subscription** and fill in:

Field	What it is
Report	Today there is one type: **Spend by team** (the deviation of a group's spend)
Frequency	*Every day*, *Weekly* (with the day of the week) or *Monthly* (with the day of the month)
Recipients	Emails separated by commas; at least one is required
Perspective and Group (optional)	To send only one team's spend: whoever subscribes to their group does not receive the others'. Today the Perspective is typed by its *id*

Worth knowing:

- **The cadence is evaluated in your tenant's timezone**: "Mondays" are your organization's Mondays, not the server's.
- **A 31st does not skip short months**: it is sent on the last day of the month.
- **It is sent only once per period**, even though the system checks every hour.
- **A report with no data is not sent.** An email with an empty table would read as "your team spent nothing", and it almost always means the ingestion has not run yet.
- **One email per recipient**: nobody sees who else receives it, and a bounce does not affect the rest.
- **If whoever created the subscription stops being active** in the tenant, the subscription is **deactivated** and stops sending.
- **Send a test now** uses the same path as the scheduled send and **does not count** as the period's send.
- The chart goes **inside the email body** (not as an attached image) so it looks the same in Gmail and Outlook.

8.7. Integrations & API (`/admin/integrations` , Enterprise)

A hub with three tabs: **Copilot M365** (tenant configuration and assisted chat over your own FinOps data), **MCP API Keys** (keys for integrations with MCP-type AI assistants: generate, rotate and revoke) and **Public API** (documented REST with OpenAPI/Swagger, with usage limits and monitoring; see 11.3).

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8.8. Billing & Audit (`/admin/account` , Professional+)

Two tabs: **Billing** (plan, quotas, payment method and plan change; see 8.4) and **Audit Trail** (the log of who changed what and when; see 11.2).

8.9. Other Administration modules

Module	Where it is	Plan	Use
Azure Lighthouse Onboarding	Users & Access → Lighthouse Onboarding	Enterprise	ARM template for cross-tenant delegation, at `/admin/onboarding/lighthouse`
AI Configuration	Configuration → AI	Professional	Enable or disable AI features, choose the model, adjust sensitivity and what data is shared
Invoicing Report	Reports & Export	Business	JSON/CSV/PBIT stub export with detail by billing profile, invoice section and customer
Workbooks	Reports & Export	Enterprise	Custom reports exportable to PDF and shareable by link
Audit	Billing & Audit → Audit Trail	Professional	Complete log of who changed what and when, filterable by user, action and date, exportable for compliance
MCP API Keys	Integrations & API	Enterprise	API keys for MCP-type integrations — generate, rotate and revoke
Public API	Integrations & API	Enterprise	Documented REST API (OpenAPI/Swagger), with rate limits and usage monitoring
Power BI Templates	Reports & Export	Enterprise	Downloadable `.pbit` templates, pre-connected to your SaaS data
FOCUS 1.1 Export	Reports & Export	Professional	Export of your data in the standard FOCUS format, schedulable for automatic daily generation
SAML SSO	Users & Access	Enterprise	You configure your Identity Provider, map attributes (roles, emails), test and enable it for the whole organization
Partner Markup (CSP)	Configuration	Enterprise	Configurable margins per service for CSP partners, applied automatically in billing
M365 Copilot	Integrations & API	Enterprise	Tenant configuration + assisted chat over your own FinOps data

8.10. Add-ons Marketplace: à-la-carte modules and capacity (`/marketplace`)

If you need a feature or a quota your plan does not include, **you do not have to change plans**: the Marketplace sells standalone add-ons. They are bought as a **prorated monthly subscription** or as a **temporary pass** of 1, 3, 6, 9 or 12 months, and paid through the same gateway (Paddle) as your plan.

What is sold:

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Type	What it includes
Features	*Commitment Simulator for RIs & Savings Plans*, *Advanced Zombies & Networking Audit*, *Executive PDF Reports* and the **navigation modules**: Resource Inventory, Databases, Compute, Networking, TTL, Power Schedules, High Availability, Expiring Credentials, Approvals, Optimization & Savings, Storage, Users & Licenses, Azure AI, Azure Integration Services, Monitoring, Security, Advanced Analytics, Policies & Auto-Block and Governance Reporting
Quotas and capacity	Additional Azure subscription, extra user seats (+5) and additional tenant

How to use it:

1. Filter by *Features* or *Quotas*. Each card explains what it unlocks and shows the current price.
2. **Temporary pass:** choose the duration and press **Activate Pass**. It is paid once and, when it expires, the module switches off by itself without touching your base plan. If you already have one active, **Extend Pass** appears.
3. **Monthly subscription: Add to monthly subscription** adds the add-on to your current subscription with proration.
4. At the top, **Your Active Capabilities and Passes** shows what you have and how many days each pass has left.

A purchased module opens **both the screen and its APIs**, and you get a notice before it expires. Activation happens when the gateway confirms the charge, a few seconds later.

The **additional tenant** —adds another Azure tenant to your contract, with your same account and accesses, inheriting your plan's limits— is purchased in **Billing → Expand capacity** (8.4.2): it modifies your existing subscription instead of opening a separate purchase, and the Marketplace shows it so you can find it.

Enterprise customers have everything included (they will see the *Enterprise Plan Active* badge). If your subscription is billed through the **Microsoft Azure Marketplace**, additional modules cannot be purchased from here: write to us and we will sort it out through your purchasing channel.

8.11. SaaS Operations (SuperAdmin) (`/superadmin/ops`)

Global operations control center for SaaS monitoring:

1. Check overall platform status and unacknowledged alerts.
2. Review component health and cron execution status (last run, age, summary, state).
3. Verify notification-channel coverage across SuperAdmin tenants.
4. Dispatch operational notifications to configured channels when degradation is detected.

8.12. SuperAdmin-only modules

The screens only the CSCloudSolutions team sees (system role **SuperAdmin**), at the end of the *Administration* group of the menu:

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Screen	Route	What it is for	Where it is explained
Tenant Management	`/superadmin/tenants`	Manual tenant creation, tier and status, salesperson, commission, sale date and term, secure impersonation	1.3 and 14
Signup Funnel	`/superadmin/funnel`	Signup conversion, active trials and onboarding progress	21
Support (Global)	`/superadmin/support`	Ticket queue of all tenants	8.1
SaaS Operations	`/superadmin/ops`	Component and cron status, operational notifications	8.11
Partner Center Alerts	`/superadmin/partner-alerts`	PAL/CPOR states per tenant and recent events	1.3
Affiliate Program	`/superadmin/affiliates`	Collaborator sign-up, referral links and the partner panel	17
Commissions	`/superadmin/comisiones`	Commission ledger, sales reps and payout runs	17.6 and 20
Platform Usage	`/superadmin/consumo`	AI tokens absorbed by the house and calls to Azure APIs	22
Pricing Units	`/superadmin/pricing-units`	Units of Measure normalizer under FOCUS 1.1	23
Load Testing	`/superadmin/load-test`	Real concurrent traffic against the server	24
AI — Global Configuration	`/superadmin/ai-global-config`	Fallback AI provider and key, master switch	25
Global Announcements	`/superadmin/announcements`	Banners and pop-ups for users	13

The rest of the platform flows —tenant lifecycle (14), Azure Marketplace (15) and onboarding via Azure Lighthouse (16)— have their own section.

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9. FinOps Copilot (AI Assistant)

Floating icon in the bottom corner of the screen, available on Professional+.

- **Automatic context awareness:** the Copilot reads the content of the page you're on — you don't need to tell it which module you're in. When you open it, without typing anything, it generates an **executive report** of what's being shown: module context, key findings, savings opportunities prioritized by impact, risks, and a 7-day action plan.
- **Targeted questions:** besides the automatic report, you can ask it directly. E.g., in Budgets: *"Summarize the current state of our budgets"*.
- **Strategic suggestions and information:** the Copilot provides exclusively analytical recommendations, cost optimization suggestions, and informative insights to support decision-making — it does not perform corrective actions or direct modifications on your infrastructure.
- **Continuous conversation:** the Copilot remembers the conversation even if you change pages, so you can ask it *"compare this rightsizing with the anomalies we just saw"* and it resolves the reference to what was discussed on

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the previous screen. It remembers the latest questions of the chat (up to 8 turns) and, if an old turn contradicts what the screen shows today, **the current figures prevail**.

- **Monthly query quota:** Professional **50**, Business **150** and Enterprise **unlimited**. If the tenant configures **its own AI key** (*Configuration* → *AI*), the Copilot has no query cap.
- **New conversation:** the button in the panel header clears the thread to start from scratch (it is disabled while the Copilot is answering).
- **Comfortable reading:** the answer arrives word by word and the panel follows it; if you scroll up to reread a report, it does not drag you back to the bottom until you ask a new question.

> The Copilot lives inside the platform: it does not appear on public pages (legal, status, partner panel, demo, login and sign-up) or before signing in.

---

10. Account Security (MFA)

TOTP-based two-factor authentication (Google Authenticator, Microsoft Authenticator, Authy, etc.), optional per user, required only for **sensitive operations** (deleting a tenant, canceling a subscription, changing billing configuration).

How to enable it:

1. Go to your profile → **Security Settings** → **"Enable 2FA"**.
2. The system shows you a QR code — scan it with your authenticator app.
3. Enter the 6-digit code shown by the app to confirm.
4. You receive **10 recovery codes** — download and store them somewhere safe. They're shown only once.

When it's requested: when you attempt an operation marked as sensitive, a modal appears asking for your 6-digit code (or a recovery code if you've lost access to the authenticator). Each recovery code is single-use; if you use up all 10, you must disable and re-enable 2FA to generate a new set.

If you lose access to your authenticator: use any of your 10 recovery codes to log in and reconfigure 2FA from scratch. If you've also lost the recovery codes, an Admin can force-disable your MFA (logged in the audit trail).

---

11. Advanced Features and Integrations

This section is for technical users (Cloud Admin, DevOps) who need to integrate the platform with other tools or configure advanced access.

11.1. SSO SAML (`/admin/sso` , Enterprise, via WorkOS)

Lets Enterprise customer users log in with their own Identity Provider (Okta, Auth0, AD FS) instead of — or alongside — Microsoft Entra ID.

Configuration per customer:

1. Go to **Admin → SSO SAML** (only visible on Enterprise tier).
2. Get the `workos_org_id` and `workos_connection_id` from the WorkOS dashboard (CSCloudSolutions creates the Organization/Connection there if they don't exist).
3. Fill in the form: **Domain** (e.g., `acme.com`), **WorkOS Organization ID**, **WorkOS Connection ID** → enable the **Enable SSO** toggle → save.
4. **"Generate Admin Portal"** button → opens a WorkOS link in a new tab; the customer's IT admin receives an email to configure their own IdP.

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5. (Optional) **"Test SSO"** button → redirects you to the IdP to validate the login works before announcing it to end users.

How the end user logs in: on the login screen, they choose "Login with SSO" instead of "Sign in with Microsoft" → the system redirects them to their own IdP → after authenticating, they're automatically returned to the dashboard.

> ⚠️ Only **one IdP connection per tenant** is supported by current design. SSO coexists with Microsoft Entra ID login (MSAL) without replacing it — both methods work simultaneously, so you don't lose existing access when enabling SSO. The SSO session lasts 12 hours.

Common errors: "SSO not configured" (platform credentials missing), "SSO not enabled for this tenant" (the toggle is off), "Missing workos_connection_id" (the portal hasn't been generated yet, or the customer hasn't configured their IdP).

11.2. Audit Trail — advanced usage (`/admin/audit`, Professional+)

Beyond the filterable log (by email, action type, status, and date range, 10 rows per page), you have 4 export formats: **CSV (current page)**, **Full filtered CSV (streaming)**, **JSON**, **NDJSON**.

Typical use cases:

- **Compliance audit (SOC2):** export the full quarterly range with the filtered export button.
- **Investigate an incident:** filter by the suspicious user's email + `FAILURE` status.
- **Operational monitoring:** filter by action type `DELETE` + a recent date range to see what was deleted.

> ⚠️ The full export has a **100,000-row cap** — for larger datasets, narrow the date range and export in parts. Log retention is currently indefinite (no automatic purge yet).

11.3. Public REST API v1 (`/admin/api-keys`, Enterprise)

Read-only programmatic access to your data (costs, budgets, recommendations, anomalies) to connect BI tools or your own scripts.

How to generate and use it:

1. **Admin → Public API** → generate a new key, choosing only the **scopes** you need (`read:cost`, `read:resources`, `read:budgets`, `read:recommendations`, `read:anomalies`, plus the write ones, `write:metrics` and `write:shift-left`, which must be granted on purpose: 11.10 and 11.11) — request the minimum necessary, e.g., `read:cost` if it's only to feed a BI dashboard.
2. Authenticate each request with the header `X-API-Key: pak_live_xxx` (or `Authorization: Bearer pak_live_xxx`).
3. Example endpoints: `GET /cost/summary?from=...&to=...&groupBy=service`, `GET /cost/timeseries?granularity=daily|monthly`, `GET /resources?type=...&limit=100`, `GET /budgets`, `GET /recommendations`, `GET /anomalies?severity=low|medium|high`.
4. Interactive documentation (Swagger) available at `/api/v1/docs`.

> ⚠️ Default limit: **60 requests/minute** per key (`X-RateLimit-*` headers show your remaining quota; hitting the limit returns a 429 error). If you need more for a batch job, request it from Admin → Public API or via support (up to 1000 req/min). Monetary values always travel as **strings** (e.g., `"1234.56"`), not numbers — parse them as text to avoid losing precision.

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11.4. Showback/Chargeback PDF Export (`/admin/report` , Business+)

Generate and email showback/chargeback invoices to each customer or internal cost center.

- **Download a single invoice:** from the billing report, select the customer/center and period → the PDF downloads directly.
- **Download all at once:** same flow without selecting a specific customer → a ZIP with all the period's invoices downloads.
- **Send via email:** fill in the recipient's email and name → it's automatically sent with the PDF attached via Microsoft 365 (requires CSCloudSolutions to have email sending configured for your tenant — if unavailable you'll see a clear error asking you to contact support).

Every send and download is logged in the audit trail.

11.5. FOCUS 1.1 Export — advanced usage

Beyond manual download from `/admin/focus-export` (see section 8.7), you can automate extraction with your own MCP API Key (generated at `/admin/mcp-keys`):

```
curl -H "Authorization: Bearer mcp_xxx" \
  ".../api/exports/focus?tenantId=...&format=ndjson"
```

Available formats: CSV, JSON, NDJSON. Cap of **500,000 rows** per export (default 100,000). Useful for automatically feeding external FinOps tools (Power BI, CloudHealth, etc.) without manual intervention.

11.6. Status Page

A public page (no login required) where you and your users can check at any time whether the platform is operational: `/es/status` (or `/en/status` , `/status`). Shows the status of API, database, Azure sync, AI provider, and billing, plus an incident history. Useful to share with your team if something seems broken — before opening a ticket, check whether an incident is already reported there.

11.7. Paddle Billing — technical detail

The plan change (see section 8.4) has 3 possible proration modes when upgrading/downgrading:

- **Immediate proration:** the difference is charged or credited right now.
- **Next-cycle proration:** the full price change is charged only at the next renewal.
- **No immediate charge:** the plan change applies without generating any charge until the next cycle.

From `/admin/billing` you also access the **payment management portal** (update card) and the **downloadable invoice history**.

11.8. 7-day free trial (self-service)

If you signed up yourself from the pricing page (not through an onboarding assisted by the CSCloudSolutions team), your account starts with a **7-day trial** on the plan you chose. You'll see a trial status banner at the top of the platform:

- 🔵 Blue: 5 or more days remaining.
- 🟡 Yellow: between 3 and 4 days remaining.
- 🔴 Red: 2 days or fewer — this one can't be dismissed.

You can upgrade to a paid plan at any time from **Billing** — the trial immediately converts to an active subscription. If the trial expires without an upgrade, the account switches to limited access until you activate a paid plan.

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11.9. Infrastructure and data residency

- **Active physical region:** Azure West US 2.
- **Production Redis:** Azure Managed Redis with HA enabled.
- **Edge/CDN:** Cloudflare is used in front of the Azure origin.

11.10. Shift-Left FinOps: API and GitHub Action (Business+)

What it does and how to read the result: see 7.10. Here, the technical part.

The permission. The `write:shift-left` scope does not come with existing keys: it must be granted explicitly when generating the key in **Integrations & API → Public API** (11.3). This is on purpose: a cost-reading key should not be able to write estimates.

The API. `POST /api/v1/shift-left/estimate`, with the API Key in the `X-API-Key` header. It receives the Terraform plan as JSON (the output of `terraform show -json`, up to 6 MB) and returns the monthly cost delta, the projected cost, the number of unpriced resources and the state of the repository's Budget Gate (`OK`, `EXCEEDED` or `NOT_EVALUATED`).

The GitHub Action *FinOps PR Cost Estimator* wraps that call in a step of the Pull Request workflow:

```
permissions:
  contents: read
  pull-requests: write   # without this it cannot comment

steps:
  - uses: actions/checkout@v4
  - uses: hashicorp/setup-terraform@v3
  - run: terraform init
  - run: terraform plan -out=tfplan.bin
  - uses: ./integrations/github-action      # the action, as you have it in your repository
    with:
      api-key: ${{ secrets.FINOPS_API_KEY }} # always via secrets, never literal
      plan-file: tfplan.bin
      threshold-usd: '200'                   # optional: local Budget Gate
```

Inputs: `api-key` (required), `plan-file` (defaults to `tfplan.bin`), `plan-json` (an already-converted plan, so Terraform is not needed on the runner), `threshold-usd`, `comment` and `api-url`. Outputs, to chain with other steps: `delta-usd`, `projected-usd`, `unpriced` and `gate-status`. It needs `jq`, `curl` and `bc`, which already ship with GitHub runners.

11.11. Business metrics ingestion through the API (revenue and volume)

To calculate the **unit cost** (Unit Economics, 5.14) and the **gross margin** (Leaderboard, 5.2), the platform needs to know how much you sold or processed. You send it from your pipeline or a cron with a public API Key that has the `write:metrics` scope:

```
curl -X POST https://<host>/api/unit-metrics/ingest \
  -H "x-api-key: pak_..." -H "Content-Type: application/json" \
  -d '[{"metricDate":"2026-08-21","metricType":"TRANSACTIONS","unitCount":14320}]'
```

Up to **400 rows per call**; the tenant comes from the key itself, never from the request body. `metricType` = `REVENUE` loads **revenue** (in money, not a unit count: it is not offered as a denominator of the unit cost) and enables the gross margin.

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11.12. Mobile app (PWA)

The platform can be installed on a phone as a **progressive web app** (add to home screen) and, on narrow screens, offers a dedicated mobile experience at `/mobile` with five tabs:

Tab	What you see
Summary	Month cost, forecast, potential savings and zombie resources
Alerts	The state of your alert rules (triggered, paused) and credentials that are expired or expiring in 30 days or less
Approve	Pending remediation approvals, with the **Approve** button to resolve them from the phone
Support	Open a ticket with the team
Profile	Your account, the language and access to the desktop version

The summary has shortcuts to **Alerts** (budgets, anomalies and credentials), **Recommendations** (the savings suggested by Azure Advisor) and **Support**. If you prefer the desktop version, choose it with **Use desktop version**.

---

12. Best Practices

- **Weekly review:** check the **Dashboard** and **Zombie Resources** at least once a week to catch financial leaks before they pile up.
- **Automate early:** enable **Power Schedules** on your Development/Testing environments as a first step — it's common to see 60% savings in idle compute hours with no other change.
- **Enforce tag compliance:** without consistent tags, the chargeback/showback module can't fairly distribute the monthly bill across teams — it's the foundation everything else relies on.
- **Use the What-If Simulator before committing:** before purchasing a Reservation or Savings Plan, simulate the scenario and save it — it gives you a concrete number to justify the decision to finance.
- **Set up at least one notification channel from day one** (Slack/Teams if your team already lives there, or email if you prefer simplicity) — budget alerts are useless if nobody sees them in time.

13. Global Communications

Where: sidebar → **Global Communications** (`/superadmin/announcements`). SuperAdmin only.

Use it to tell tenants something from inside the platform: scheduled maintenance, an incident, a change of terms.

13.1 Creating an announcement

Press **New announcement** and fill in:

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Field	Notes
Title and message	The message supports Markdown (bold, lists, links)
Severity	Info / Maintenance / Warning / Critical — sets the banner color
Channels	**Banner** (top strip), **Popup** (window), or both
Validity	From / to. The notice appears and disappears on its own within that window
Scope	All tenants, or a selection
Action link	Optional. Adds a "More information" link

Save draft leaves it ready without publishing; **Publish** puts it into effect according to its dates.

13.2 Specific scope

Unchecking *"All tenants"* opens a searchable tenant list — filter by name or ID, each showing its status (ACTIVE, TRIAL, PAST_DUE).

CANCELED tenants don't appear: their users no longer sign in, so targeting them has no effect. TRIAL and PAST_DUE do appear — a PAST_DUE is often exactly who you want to notify.

If you edit an old announcement targeting a tenant no longer on the list, it shows as **"Not in list"**. That's deliberate: you can uncheck it instead of silently losing it on save.

13.3 Translations (optional)

The form has a collapsible **Translations** section with fields for English and Portuguese.

They are not mandatory. If you only fill in Spanish, users in other languages see the notice in Spanish. That's deliberate: an outage notice shouldn't wait for a translation.

A half-finished translation (title only, or message only) is **discarded on save** — showing a blank body would be worse than showing the full text in another language.

The **Languages** column shows at a glance which ones each announcement has.

13.4 Statuses

Status is computed from the dates:

Status	Meaning
Draft	Saved, not published
Scheduled	Published, start date not reached yet
Active	Showing right now
Finished	Past its end date
Cancelled	Manually withdrawn

Cancel (≣) pulls it out of circulation immediately without deleting it. **Delete** (▨) removes it along with its dismissal records; it cannot be undone.

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14. Tenant Lifecycle

The **Tenants** view has three new columns, hidden by default — enable them from the column picker:

- **Activation Date** — when it was effectively activated
- **Cancellation Date** — when it was cancelled, if applicable
- **Cancellation Reason** — Voluntary churn / Payment delinquency / Contract expired / Admin deprovisioning

These dates are now recorded automatically: when a cancellation arrives from the payment gateway or the marketplace, when a contract expires, or when an account is suspended for non-payment. A **full history** of every transition is also kept, so a tenant that cancelled and came back retains all its periods — which is what cohort retention analysis needs.

> **About tenants created before September 2026:** their "Activation Date" is the record creation date, not the effective onboarding or first payment. That distinction wasn't recorded at the time and cannot be recovered. Treat them as approximate.

14.1 Filtering and export

Above the table you can pick **Activations** or **Cancellations** and set a date range. The filter stacks with the search box, so you can combine both.

Export CSV downloads exactly what you're seeing — filters applied — including a **tenure in months** column (activation → cancellation, or activation → today if still active). It's built for the accounting report: opens correctly in Excel, with proper accents.

15. Azure Marketplace

The platform can also be sold through the Azure Marketplace. The technical flow is already built; what's left to publish is in `docs/marketplace-publicacion-checklist.md`.

15.1 What happens on purchase

1. Azure sends the buyer to the landing page with a temporary token.
2. The platform resolves that token against Microsoft and shows the purchased plan.
3. The customer signs in with Entra ID and activates.
4. Microsoft starts billing and notifies us by webhook of every change: plan change, quantity, suspension, reinstatement and cancellation.

15.2 Two statuses that are NOT the same

The tenant record shows two similar fields, and the difference matters:

Field	What it is
Marketplace status	What **Microsoft** says: Subscribed, Suspended, Unsubscribed
Subscription status	**Our** commercial status, also driven by Paddle and your actions in this panel

They're kept apart deliberately. As a single field, a suspension originating in Azure would erase the real reason for a cancellation handled through another channel — and in a billing dispute you couldn't explain where each status came from.

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15.3 Reconciliation data

For each Marketplace subscription we store the offer, the purchaser's email and the directory they purchased from. **The purchaser isn't always the person who later uses the platform**, and that distinction is what Microsoft asks for to resolve a dispute.

---

16. Onboarding via Azure Lighthouse (Enterprise)

This is the alternative to the onboarding script: the customer **delegates** access to their subscriptions from their own directory, without handing you credentials or running anything in their environment. It lives under **Users & Access → Lighthouse Onboarding**, and only appears on the **Enterprise** plan.

16.1 When it fits and when it doesn't

Good fit	Poor fit
The customer won't run scripts in their environment	You need to read secrets from their Key Vault or blobs from their storage
The customer wants to be able to revoke access at any time	The customer manages spend by management group
You'd rather not maintain one secret per customer	The customer has a single subscription and already ran the script

Lighthouse delegates the **control plane only**. There is no access to the customer's data plane, and their management groups are not delegated — delegation is per subscription.

16.2 The procedure

1. **Fill in both fields.** *Managed Tenant ID* is the GUID of the customer's directory; *Managed Subscription ID* is the GUID of one of their subscriptions.
2. **Pick the roles.** For a visibility-only onboarding, **Reader** and **Cost Management Reader** are enough. *Contributor* grants write access over every resource in the subscription: ask for it only if you are going to operate the customer's infrastructure.
3. **Generate the ARM template** and download or copy it.
4. **The customer deploys it** in the Azure portal, inside **their own directory**: *Deploy a custom template → Build your own template → paste the JSON*. They need to be Owner of that subscription.
5. **Verify the delegation** with the button in the panel.

16.3 Step 5 is not optional

Generating the template and having the customer deploy it are two different things. The row stays at **"Pending acceptance — Not confirmed in Azure"** until you verify, and until that moment the platform **keeps accessing through the previous model**: the delegation exists in Azure and goes unused.

Verify delegation asks Azure, and it is the only thing that changes the tenant's access mode. If it reports active, open the customer's cost dashboards to confirm data is arriving.

16.4 A customer with several subscriptions

You don't need to generate one template per subscription. The same template delegates *whichever subscription it is deployed into*, so the customer applies it once in each one, switching context between deployments.

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> The *Managed Subscription ID* field **does not target the template**: it only feeds our record. If the customer deploys it into a different subscription, that other one is what gets delegated. That is what makes the template reusable, but it's worth knowing.

Delegated subscriptions show up on their own at the next sync cycle; there is no need to register them by hand.

16.5 The two rows and the Delete button

Once a delegation is active you will see **two** entries for the same subscription: one reported by Azure and one that is our own record. That is expected.

Delete removes **our record**, not the delegation. The `registrationAssignment` lives in the customer's subscription and can only be removed from their directory. If you delete one that was still active, the warning tells you so: the customer keeps delegating access that the platform no longer records.

For the rows Azure reports, the button rejects them with the reason — they are not our records.

> If you remove the **last** delegation for a tenant, it automatically reverts to the App Registration model. That is deliberate: with no delegation behind it, a tenant in Lighthouse mode would be left with no route into Azure at all.

16.6 If something doesn't work

Message	What's going on
"AZURE_LIGHTHOUSE_PRINCIPAL_ID is not configured"	Platform configuration is incomplete. Without it the template would deploy successfully and grant access to nobody: that's why it is blocked beforehand. Escalate to infrastructure.
"A subscription cannot be delegated to the directory it already belongs to"	The *Managed Tenant ID* is ours. Lighthouse exists so that the **customer's** directory delegates into ours.
"Available on the Enterprise plan only"	The tenant isn't Enterprise. On Professional and Business, onboarding is done with the script.
Delegation active but no data	Check the delegated roles: without *Cost Management Reader*, no consumption data arrives.

17. Affiliate Program

Path: `/superadmin/affiliates` · **Menu:** Administration → Affiliate Program

An affiliate is a third party — a reseller, influencer or partner — who promotes the platform with their own link and earns a percentage of what the customers they brought in pay.

An affiliate **is not a user of any tenant** and has no account on the platform: they get their own read-only dashboard through a personal link that you generate (17.5). There they see their referred customers and their commissions, and nothing else — not the cloud spend of the customers they brought in, and nothing about other affiliates.

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Who does what, at a glance

You, here	The partner, on their own
Register them and set their %	Shares their referral link
Generate their dashboard link	Opens the dashboard to see their numbers
Build the run and transfer (17.6)	Invoices you, where applicable

17.1 Registering a partner

1. **New affiliate**.
2. Fill in name, email and **referral code**. The code goes in the URL, so it accepts letters, numbers, hyphens and underscores, and must start with a letter or number. It has to be unique.
3. **Commission (%)** — 20 % by default. This is the percentage applied to every charge made to that affiliate's customers. **Mind the standard program:** a referral is paid **15 %** (co-selling, 20 %, is registered as a sales rep: section 20), so **adjust the percentage when signing them up**.
4. *Payout method* and *Payout reference* are free text for whoever settles payments (bank transfer, PayPal, alias). No structured bank details are stored.
5. **Save**.

If the email or the code already exists, the creation is rejected with a notice: both are unique on purpose, so two affiliates cannot claim the same attribution.

17.2 The referral link

Copy on the affiliate's row puts this on the clipboard:

```
https://finops.cscloudsolutions.com.ar/signup?ref=CODE
```

When someone arrives through that link, the code is stored in their browser for **60 days**. If they sign up within that window, the customer is attributed to that affiliate.

Three rules worth knowing when answering questions:

- **First touch wins.** A customer belongs to exactly one affiliate and is never reassigned. If they came through A's link and later through B's, the customer is A's.
- **No self-referrals.** If the affiliate's email matches a user of the customer, the attribution is rejected.
- Attribution happens **when the customer signs up**. A customer that already existed is not attributed by visiting a link later on.

17.3 Editing, suspending and deleting

The pencil button opens the same form used for creation, pre-filled. You can change name, email, code, percentage, status and payout details.

Two things worth understanding before you touch anything:

- **Changing the percentage does not rewrite history.** Every commission stores its own snapshot of the percentage in force when it accrued. The new value applies from the next charge onwards.
- **Changing the code breaks links already shared.** The visitor's cookie stores the code, not the affiliate, so anyone arriving through the old one is left unattributed. The form warns you when you are changing it.

Suspend is what to use when you stop working with someone: it stops new commissions from accruing and keeps the entire history. The affiliate stays in the list, greyed out, with their numbers intact. **Reactivate** puts them back.

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Delete only works if the affiliate has **no commissions at all**. If they have even one, pending included, the platform refuses and tells you how many there are. This is not an arbitrary restriction: given how the database is built, deleting the affiliate would take their commission history with it — including what you already paid them — and that record cannot be rebuilt. That is what *Suspend* is for.

If the affiliate has referred customers but no commissions yet, they can be deleted; the confirmation notice tells you how many customers will be left unattributed, and that cannot be undone.

17.4 Commissions and settlement

The **Accrued commissions** table fills itself: every time an attributed customer pays, a commission is recorded.

Through each customer's first year, and no further. The window is 12 months from that customer's first charge: a monthly customer accrues their twelve charges, and an annual one gets the initial charge while next year's renewal falls outside. It is measured in months rather than in number of charges precisely for that reason — "up to 12 charges" would give an annual customer twelve years.

The partner dashboard shows the cut-off date per customer, so "how long do I earn on this one?" answers itself.

Status	What it means
Pending	Accrued. Just charged to the customer; it is worth waiting out the refund window before approving.
Approved	Reviewed and ready to pay.
Paid	Already settled with the affiliate.
Reversed	The customer requested a refund and the commission was voided automatically.
Cancelled	Voided manually.

To settle: tick the rows, click **Approve**, and once the transfer goes out, **Mark as paid**.

> A **Paid** commission cannot be rolled back from the interface, and a later refund will not reverse it either. If it needs correcting, deduct it from the next settlement: that is an accounting operation, not a click.

The **Unsettled** and **Paid** columns in the partners table are totals in the currency of the most recent charge. If one affiliate has customers paying in different currencies, those totals are not comparable — check the per-commission detail.

17.5 The partner dashboard

The **dashboard** button on the affiliate's row generates their personal link:

```
https://finops.cscloudsolutions.com.ar/en/partner#k=afp_live_...
```

It is shown only once. The database keeps nothing but its hash, so there is no way to see it again: if you lose it, you issue a new one — and that invalidates the previous link. Same deal as API keys.

What the partner sees: their totals (pending and already paid), their referral link, their referred customers with the date each one stops earning, and the detail of their commissions with each status. From the same panel they download the **Partner manual** (PDF or Markdown, in Spanish, English and Portuguese): the material to understand the platform and sell it.

What they do **not** see, worth knowing so you can answer: the cloud spend of the customers they brought in, their Azure identifiers, or anything about other affiliates. Only the company name, the dates and their own commission amounts.

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The **broken-link** button revokes access without touching the affiliate or their commissions. And **suspending** an affiliate also closes their dashboard: there is no need to revoke as well.

> The link **is** the credential: whoever holds it sees those numbers. Send it to the person over a direct channel, do not paste it into a group chat.

17.6 Payout runs

Path: `/superadmin/comisiones`

A run groups everything due in **one currency** into a batch, per beneficiary, and is what ties a transfer to the commissions it covered. Affiliates and sales reps go together: they are in the same ledger.

The monthly cycle:

1. **Pick a currency and a per-person minimum** and build the run. The minimum applies to **each beneficiary's total**, not to each commission: twelve commissions of $0.30 are $3.60, which may be worth a transfer even though none of the twelve is on its own. Anyone below it stays out and their balance carries over to the next run.
2. **Download the payment order** (CSV). One row per person with the amount, email, payout method and payout details: what you need in order to transfer, without looking up each record.
3. **Transfer**, by whatever means you agreed with each of them.
4. **Confirm the payment** with the reference. Only then do that run's commissions become *Paid*, all together and under the same reference.

Cancelling an open run releases its commissions for the next one. A run already paid cannot be cancelled: the money left, and that is fixed with an adjustment.

Three things that prevent surprises:

- **One currency per run.** You cannot transfer in two currencies at once, and the platform does not invent an exchange rate.
- **Two runs do not collide.** Commissions taken by one are reserved; a run built in parallel will not take them.
- **Confirming twice does nothing.** It is safe to retry if the connection dropped.

The minimum you used is stored on the run. Changing the default next month does not rewrite why an old run left someone out.

---

18. Spend anomalies: who handles them

When the platform detects unusual spend, it no longer stops at the alert.

18.1 The owner

Each anomaly shows **who is responsible** and where that assignment came from. The platform derives it from your own governance model, in this order:

1. If the Resource Group is manually assigned to a Cost Group, that group's owner wins.
2. If it matches a Cost Group's name pattern rule.
3. If it carries the tag that defines that group.
4. If the resource has an `Owner` tag with an email address.

If none of those resolve, you'll see "Unassigned" — deliberately. We'd rather tell you an owner is missing than assign it to the wrong person: they'd learn to ignore the alerts. That "unassigned" count is, in practice, your list of what still needs tagging.

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> It's recalculated on every scan. If you assign the Resource Group to a Cost Group **after** the anomaly fired, it finds its owner on its own.

18.2 Follow-up

Each anomaly has a status that **is saved**, along with who changed it and when:

Status	When to use it
New	Nobody has looked at it yet
Investigating	Someone picked it up
Resolved	Acted on and spend corrected
False positive	The detection was wrong

That last one isn't the same as "dismissed": it's feedback we use to tune detection.

Changes are visible to your whole team, so two people don't work the same item unknowingly.

18.3 The alert

If the anomaly has an owner, the notification **goes to that person**. If it doesn't, it goes to the whole team — an unowned deviation still needs someone to see it.

---

19. Subscriptions beyond your plan's limit

Configuration → Account Status (formerly "Cloud Accounts") lists **every** Azure subscription the platform discovers. If you have more than your plan allows (Professional 2, Business 3, Enterprise unlimited), the ones left outside appear **dimmed** with an amber **"Over plan limit"** badge.

That badge means something specific: **the subscription is visible in this table and feeds no dashboard**. Its spend isn't counted in the totals, its resources don't appear in the inventory, and its anomalies aren't detected.

19.1 Choosing which ones are monitored

The platform can't know which ones matter to you, so it takes the first ones by identifier order — a stable but arbitrary criterion. You have two ways to decide for yourself:

- **Unlink the ones you don't care about.** The *Unlink* button on each row takes a subscription out of the count and **frees up the slot** for another one. It asks you to type the identifier to confirm, because the action affects every dashboard in the tenant.
- **Increase the quota.** The *Increase quota* button in the banner adds subscriptions without changing plans. What you buy is kept even if you switch tiers later.

19.2 Relinking

An unlinked subscription doesn't disappear: it moves to the end of the list, dimmed, with an **"Unlinked"** badge and a **Relink** button.

Unlike unlinking, relinking **doesn't ask for confirmation**: it destroys nothing — it only lifts the exclusion — and unlinking again is one click away.

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> Relinking doesn't require redoing the onboarding or running any script again. The historical spend of an unlinked subscription **is preserved**: closed months are accounting data and are never deleted.

> If you are at your plan's limit, relinking brings the subscription back into this table but the dashboards still leave it out, the same as any other one above the limit. Make room first.

19.3 Adding a new subscription in Azure

You don't need to redo the onboarding. The script assigns the read roles at the root management group of your directory, and Azure inherits them downward: a subscription created later is born with permission already. It shows up on its own with **Sync now**.

The only manual step is if you want to **act on it** — shutting down VMs, power schedules, right-sizing: the remediation role is assigned per subscription and is not inherited. Azure portal → the subscription → Access control (IAM) → Add role assignment, on the same Service Principal.

> If the new subscription **doesn't** appear after syncing, the assignment at the root management group never happened when the onboarding was run. In that case the read roles have to be assigned on that subscription.

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20. Sales commissions

Route: `/superadmin/comisiones` · **Menu:** Administration → Commissions

A **sales rep** is whoever **closes the sale** of a tenant. This is not the same as an affiliate (17), who only brings the client in with their link: the sales rep earns a percentage of what the client they sold pays, with their own accrual schedule. Sales-rep and affiliate commissions live in the **same ledger** and are settled together in the payout runs (17.6).

> **Who does what, at a glance**
>
> | You, here | The system, on its own |
> |---|---|
> | You sign up the sales rep and load the sale on the tenant | It accrues the commission on every charge that qualifies |
> | You attribute the sale to the sales rep | It calculates the amount using the percentage snapshot of that moment |
> | You build the run and confirm the payment | It cancels the commission if the client asks for a refund before it is paid |

20.1 Signing up a sales rep

In the **New sales rep** card fill in first and last name, email and the **commission percentage** (20% is the default of the agreement) and press **Create sales rep**. The list shows how many sales each one has.

20.2 Recording the sale on the tenant

A commission **does not accrue without three pieces of data on the tenant**, shown as columns in **Tenant Management** (`/superadmin/tenants`):

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- **Salesperson** and **Commission (%)**, editable. If a sale was closed with a condition different from the general one, the agreement's percentage prevails over the sales rep's.
- **Sale date:** where accrual starts from. For tenants loaded before this feature it shows *not recorded* (in amber): **a retroactive date is not invented**, because a commission calculated on a guessed date is worse than not calculating it. Once recorded it **is not corrected from this screen**: moving it would shift the start of accrual and recalculate commissions already settled.
- **Contract term:** *annual* or *monthly*. It decides which schedule applies.

With that loaded, in the **Attribute a sale** card type the **tenant ID**, pick the sales rep and press **Attribute**. Without a sale date nothing accrues.

20.3 How much and when they earn

The reference commission is **20% of the annual value of the sale**. The two modalities come from that:

Contract	How it accrues
Annual	A single installment: 20% of the annual charge, **due from the 2nd month** after the sale (the retention period). The following year's renewal does **not** pay it again.
Monthly	One installment per month of **1/12 of the annual 20%**, which equals **20% of each monthly charge** (the annual value is the charge × 12). **Month 1 is not settled** (onboarding grace) and falls due together with month 2: after the second charge the first two installments are settled. **From month 3 on** every successful charge accrues its installment and is due immediately, up to **12 installments**: month 13 accrues nothing.

Rules worth knowing in order to answer questions:

- **If the client leaves, the installments that were not yet due are cancelled.** Cancelling before month 2 settles nothing; cancelling in month 4 stops the future installments, and whatever was already due or paid is **left untouched**: deducting from someone a commission they already earned is an accounting decision, not an automatic side effect.
- **A refund cancels the commission** as long as it is not paid. If it was already paid, it is deducted from the next settlement (17.4).
- **It does not accrue** if the tenant has no commercial agreement (direct or self-service sale, which is the normal case), has no sales rep assigned, the sales rep is not active, or the sale has no date. They are not errors: they are sales without a sales rep.
- **Months do not overflow:** a sale on January 31 falls due a month later on February 28 or 29, not on March 2 or 3.
- Amounts are calculated with decimal precision; they are amounts paid to a person.

20.4 The ledger

The **Ledger** table shows each movement with tenant (client), installment, commission, status and due date; it is filtered by beneficiary type (*Affiliate* or *Sales rep*) and by status. The **What each person is owed** summary separates what is **Due**, what is **In retention** and what is **Paid**. The statuses are those of 17.4. **Mark as paid** and the **CSV payment order** work the same as for affiliates, and the recommended way to pay is the **payout run** (17.6).

`DUE` (due) does not depend on a cron: it is calculated when the panel opens, from each commission's due date.

---

21. Signup funnel

Route: `/superadmin/funnel` · **Menu:** Administration → Signup Funnel

How self-service signups convert, and at which point of onboarding each new client gets stuck.

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The indicators at the top: Signups (30d), **Active Trials**, **Converted** (clients that moved to a paid plan), **Conversion Rate** and **Churn Rate**.

Conversion Funnel. The chart shows how many signups reach each stage, from signup to conversion. It is the reading that tells you whether the problem is at the top (few people sign up) or in the middle (they sign up and do not activate).

Recent Signups (last 90 days). A table with the email, the tenant, the **assigned plan** and the **status** (*Active*, *Expired* or *Converted*). It is filtered by email, plan and status, and paginated.

View Onboarding. The button on each row opens that client's **progression detail**, with the signup date and the three **cloud activation milestones**:

1. **App registration in Microsoft Entra ID** — the client's App Registration already exists.
2. **Reader permissions on Azure subscriptions** — the Service Principal can already read their subscriptions.
3. **First FinOps report and cost telemetry generated** — the platform has already brought in costs and shown the first result.

How to use it. A client who signed up but did not get past milestone 1 probably did not complete the Azure onboarding (1.3); one who reached milestone 2 and not 3 has the permission but did not complete the first sync, or lacks the cost roles. It is the list of whom to write to before their 7-day trial ends.

The 7-day free trial is activated by going through the checkout (11.8). Each tenant's status is moved by the payment gateway's webhook (the conversion to a paid plan) and a daily cron (the trial expiry); see also 14.

22. Platform usage

Route: `/superadmin/consumo` · **Menu:** Administration → Platform Usage

What it costs to operate: the AI tokens the house absorbs and the calls to Azure APIs. It is a panel for watching and containing costs, not a billing basis.

22.1 AI usage

It shows the period (last N days or hours) and separates who pays:

- **Tokens absorbed by the platform** (`source = platform`): paid by the house. **Today they are not billed to the client**: it is an operating cost to watch.
- **Tokens on customer key** (`source = byok`): the tenant uses its own API key, so the tenant pays and there is nothing to charge.

The detail is shown **by tenant** (calls, input and output tokens) and **by feature and model**. Demonstration tenants appear with a **demo** badge: their tokens are real —the Copilot and the executive report call the provider even if the tenant is synthetic— and the house pays them, but they belong to no client.

If the cost spikes because of an incident with the provider, the AI **master switch** (25) turns AI off for all tenants.

22.2 Azure API calls

It counts the calls in the limiter that all queries to Azure go through. **The throttling percentage is the actionable number**: a few stray 429s say nothing without knowing how many calls they were out of. The table brings, per service, the calls, the **throttled (429)**, the **% throttled** and the **time waiting**.

By tenant. Cost Management's limit is **per subscription and shared**: a single client with many subscriptions can slow down the rest. This view shows who is using the quota.

If *"No calls recorded yet"* appears, it is normal after a deploy: telemetry starts with the first sweep after it.

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23. Pricing units

Route: `/superadmin/pricing-units` · **Menu:** Administration → Pricing Units

The **Pricing Units Normalizer** is a curated table of Azure Billing *Units of Measure* (Hours, GB, Transactions, Tokens, etc.) used to **normalize aggregates** under the **FOCUS 1.1** specification: without it, "100 Hours" and "1M Tokens" cannot be compared or added up properly.

- **Catalog** (with its number of UoMs): searched by UoM, base unit or category, and paginated.
- **Test normalization:** a live test box: you type a Unit of Measure (for example `100 Hours` or `1M Tokens`) and a quantity, and see how it normalizes.
- **Edit rule:** the pencil on each row opens *Edit FOCUS 1.1 Normalization Rule*, where, among other things, the **unit name for display** is adjusted.
- **Reseed Catalog:** reloads the base catalog and reports how many units were updated.

It is a technical screen: you touch it when a new Unit of Measure shows up in Azure billing that the catalog does not know.

24. Load testing

Route: `/superadmin/load-test` · **Menu:** Administration → Load Testing

It generates **real concurrent HTTP traffic against the server itself** to measure **P95 latency** and **throughput** under load. Only the SuperAdmin sees it.

> ⚠️ **This runs real traffic against production**, not an isolated simulator. Start with conservative values. The hard limits are **maximum concurrency 50** and **maximum duration 15 seconds**.

How to use it: choose the **target endpoint**, the **concurrency** (1 to 50) and the **duration** (up to 15 s) and press **Run Load Test**. Depending on what you want to measure:

Endpoint	What it measures
`/api/health`	No database: the processor ceiling
`/api/status`	With database: the real MySQL ceiling
`/api/loadtest/probe`	Authenticated with the Service Principal: database + Redis
`/api/v1/costs`	A cost query
`/api/exports/powerbi-feed`	The Power BI feed

The **history** keeps the previous runs so you can compare before and after a change.

25. AI — Global configuration

Route: `/superadmin/ai-global-config` · **Menu:** Administration → AI — Global Configuration

The AI provider and key used **as a fallback** for any tenant that has not configured its own (per-tenant BYOK, in *Configuration → AI*). It feeds the **FinOps Copilot** and the **AI-powered executive reports**. Only the SuperAdmin sees it.

- **Master switch** *"Enable AI features (platform)"*: if turned off, AI is disabled for **all** tenants, regardless of their own switch. It is meant for incidents with the provider or runaway costs.

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- **Provider:** Google Gemini (Flash, free on the free tier), OpenAI (GPT-4o), **Azure AI** (Azure OpenAI), Anthropic (Claude) or DeepSeek. The configuration is **split by plan**: *Non-Enterprise plans* (the fallback for Professional and Business) and *Enterprise plans* (the exclusive provider for that plan). With Azure AI you need the **API Key**, the **Endpoint URL** and the **Deployment** (model).
- **API Key:** stored **encrypted**. **Test connection** makes a real call to the saved provider. **Delete API Key** leaves tenants without their own key **without AI** until you load a new one.
- **Defaults for new tenants:** the **anomaly-detection sensitivity** (*Low*, *Medium*, *High* or *Strict*, with Z-score 1.0) and **what data is shared** with the provider (resource and resource-group names, and tags). They are assigned when each tenant is created; afterwards each tenant adjusts them in its own AI configuration.

The consumption generated by this key shows up in **Platform Usage** (22).

Support and Contact

For any additional assistance, open a ticket from **Support** (`/support`) within the platform, or write to soporte@cscloudsolutions.com.ar.

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